



VALIDATION REPORT

for the CDM Project Activity

Qingdao Longxin Wind Power Project Phase I

in
P.R. China

Report No. 01 997 9105065547

Version No. 01.2, 2012-07-16

TÜV Rheinland (China) Ltd.

I. Project description:

Project title:	Qingdao Longxin Wind Power Project Phase I
Host Country:	P.R.China
Methodology: ACM0002/ Version 12.2.0	☒ Large Scale ☐ Small Scale
Annual average emission reductions (estimate):	92,183 tCO ₂ e/yr

GHG reducing measure/technology: Wind Power

Party	Project Participants	Party considered a project participant
(Host) P.R. China	China Resources Wind Power (Qingdao) Co., Ltd.	Yes

II. Validation:
Contract party: China Resources Wind Power (Qingdao) Co., Ltd.
Validation Team:

Validation Team			Role							
Full name	Affiliation TÜV Rheinland	Appointed for Sectoral Scopes (Technical Areas)	Team leader	Acting Team Leader	Local Expert	Team Member (Auditor)	Technicia Expert	Acting Tech. Expert	Trainee Auditor	Technical Reviewer
Mr. Minglong Huang	China	1.2, 5.1, 11.1, 12.1	X		X	X	X			
Mr. Lixin Li	China	1.1, 1.2, 3.1								X

Validation Phases:

- ☒ Desk Review
☒ Follow up interviews
☒ Resolution of outstanding issues

Validation Status:

- ☐ Corrective Actions / Clarifications Requested
☒ Full Approval and Submission for Registration
☐ Rejected

III. Validation Report:

Report No.01 997 9105065547	Current revision No.: 01.2	Date of current revision: 2012-07-16	Date of first issue: 2012-02-18
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Final approval: ☒	Released on: Date: 2012-07-16 By: Mr. Praveen N. Urs	Designated Operational Entity (DOE): TÜV Rheinland (China) Ltd. Unit 707, AVIC Building, No.10B, Central Road, East 3rd Ring Road, Chaoyang District, Tel.: +86 10 6566 6667, E-mail: ghg-doe@bj.chn.tuv.com
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Executive Summary – Validation Opinion

The validation team assigned by the DOE (TÜV Rheinland (China) Ltd.) has performed a validation of the “Qingdao Longxin Wind Power Project Phase I” in P.R. China on the basis of UNFCCC criteria for Large Scale Clean Development Mechanism (CDM) projects which is approved methodologies ACM0002 “Consolidated baseline methodology for grid-connected electricity generation from renewable sources” Version 12.2.0, EB58. The validation report and the validation protocol are summarizing the findings of the validation.

The review of the project design documentation and the subsequent follow-up interviews have provided DOE with sufficient evidence to determine the fulfilment of stated criteria.

The validation was executed in the following steps:

- Receipt and desk review of PDD (GSP Version: version 1, 12th July 2010)
- Public stakeholder comment process (10th September 2011 to 9th October 2011)
- On-site visit with stakeholder interviews (10th October 2011 to 11th October 2011)
- Issue of checklist with corrective action requests (CARs) and clarification requests (CLs) and the draft validation report & protocol (23rd Nov. 2011)
- Desk review of revised PDD (version 03, 18th April 2011)
- Review of proposed corrections and clarifications
- Issue of the final validation report & protocol

The host country is P.R. China. The project activity is unilateral CDM-project, the Letter of Approval (LoA) issued by China’s DNA – National Development & Reform Commission (NDRC), confirms the voluntary participation of China Resources Wind Power (Qingdao) Co., Ltd. (as a project participant), and the project assists P.R. China in achieving sustainable development.

The validation team did not observe any information that indicates that the project can be seen as a diversion of ODA funding towards China.

The project in PDD/Version 03 applies ACM0002/Version 12.2.0 – “Consolidated baseline methodology for grid-connected electricity generation from renewable sources”, in connection with the “Tool to calculate the emission factor for an electricity system (version 02.2.0)” and “Tool for the demonstration and assessment of additionality” Version 05.2.1.

The total emission reductions from the project are estimated to be on the average 92,183 tCO₂e per year over the selected 1st renewable 7-year crediting period. The calculation is based on the best available information at the time of submission to DOE for GSP and is confirmed to be correct.

The CDM monitoring plan and procedures are available and demonstrated in PDD in accordance with monitoring methodology ACM0002/Version 12.2.0 and actual situation. Relevant documents, e.g. CDM monitoring training plan, have been checked. Training programs for daily wind farm operation, maintenance and emergency preparedness have also been fully checked.

By utilization of wind power to generate electricity, the project results in reductions of CO₂ emissions that are real, measurable and give long-term benefits to the mitigation of climate change. It is demonstrated that the project is not a likely baseline scenario. Emission reductions attributable to the project are hence additional to any that would occur in the absence of the project activity.

In summary, the project proponent has resolved all Corrective Action Requests and Clarification Requests as stated in the Validation Protocol, which has resulted in an updated PDD (Version 03). Thus it is the validation team's opinion that the Qingdao Longxin Wind Power Project Phase I, as described in the PDD of 18th April 2011, meets all relevant UNFCCC requirements for the CDM and all relevant host country criteria and correctly applies the baseline and monitoring methodology ACM0002/Version 12.2.0; the validation team thus recommends the proposed project to be registered as a CDM project activity.

Abbreviations

BE	Baseline Emission
BM	Build Margin
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CER	Certified Emission Reduction
CL	Clarification Request
CM	Combined Margin
CO ₂	Carbon Dioxide
CO ₂ e	Carbon Dioxide Equivalent
CRWQC	China Resources Wind Power (Qingdao) Co., Ltd.
DNA	Designated National Authority
DOE	Designated Operational Entity
DR	Document Review
EB	Executive Board
ECCL	Easy Carbon Consultancy Co., Ltd
EIA	Environmental Impact Assessment
ER	Emission Reduction
ERPA	Emission Reduction Purchase Agreement
EPB	Environmental Protection Bureau
FAR	Forward Action Request
FSR	Feasibility Study Report
GHG	Greenhouse Gas
GWh	Giga Watt Hours
I	Interview
IETA	International Emissions Trading Association
IPCC	Intergovernmental Panel on Climate Change
IRR	Internal Rate of Return
kW	Kilo Watt
kWh	Kilo Watt Hours
LoA	Letter of Approval
LoI	Letter of Intent
L _y	Leakage
LRB	Land Resource Bureau
MoV	Means of Validation
MW	Mega Watt
MWh	Mega Watt Hours
NDRC	National Development and Reform Commission
NGO	Non Government Organisation
NO _x	Nitrogen Oxides
ODA	Official Development Assistance
OM	Operating Margin
OSV	On Site Visit
PDD	Project Design Document
PDR	Preliminary Design Report
PE	Project Emission
PPA	Power Purchase Contract/ Agreement
P. R. China	People's Republic of China

SA	Sensitivity Analysis
SO ₂	Sulphur Dioxide
SD	Sustainable Development
SPIDL	Shandong Provincial Industry Design Institute Co., Ltd
t	Tonne
UNFCCC	United Nations Framework Convention on Climate Change
VAT	Value-Added Tax

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1 INTRODUCTION

China Resources Wind Power (Qingdao) Co., Ltd. has commissioned the DOE TÜV Rheinland (China) Ltd. to perform a validation of the CDM Project Activity “Qingdao Longxin Wind Power Project Phase I” in P.R. China (hereafter called “the project”). This report summarises the findings of the validation of the project, performed on the basis of UNFCCC criteria for the CDM, as well as criteria given to provide for consistent project operations, monitoring and reporting. The term “UNFCCC criteria” refers to Article 12 of the Kyoto Protocol, the applied CDM methodologies and the subsequent decisions by the CDM Executive Board.

1.1 Objective

The purpose of a validation is to have an independent third party assess the project design. In particular, the project's baseline, monitoring plan, and the project's compliance with relevant UNFCCC and host Party criteria are validated in order to confirm that the project design, as documented, is sound and reasonable and meets the identified criteria. Validation is a requirement for all CDM projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of certified emission reductions (CERs).

1.2 Scope

The validation scope is defined as an independent and objective review of the project design document (PDD). The PDD is reviewed against the relevant criteria (see above) and decisions by the CDM Executive Board, including the approved baseline and monitoring methodology. The validation team has, based on the recommendations in the Validation and Verification Manual employed a rules-based approach, focusing on the identification of significant risks for project implementation and the generation of CERs.

The validation is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

2 METHODOLOGY

The validation consists of the following three phases:

- I a desk review of the project design documents
- II public stakeholder comment process
- III on-site visit and follow-up interviews with project stakeholders
- IV the resolution of outstanding issues and the issuance of the final validation report and opinion.

The following sections outline each step in more detail.

2.1 Desk Review of the Project Design Documentation

Table 1: The following table outlines the documentation reviewed during the validation:

/1/	UNFCCC, Clean Development Mechanism Validation and Verification Manual (VVM), Version 02.0, EB 65 Annex 4	
/2/	UNFCCC, CDM-PDD – Project Design Document form, Version 03.2, EB25, Annex 15	
/3/	UNFCCC, Guidelines for completing the Project Design Document (CDM-PDD) and the Proposed new baseline and monitoring methodologies (CDM-NM), Version 07, EB41, Annex 12	
/4/	UNFCCC, ACM0002– “ Consolidated baseline methodology for grid-connected electricity generation from renewable sources”, Version 12.2.0, EB 65 Annex 16	
/5/	UNFCCC, “Tool to calculate the emission factor for an electricity system”, Version 02.2.0, EB 61 Annex 12	
/6/	UNFCCC, “Tool for the demonstration and assessment of additionality”, Version 05.2.1, EB39, Annex 10	
/7/	UNFCCC, “Guidance on the Assessment of Investment Analysis”, Version 05, EB 62 Annex 5	
/8/	UNFCCC, Guidelines on the Demonstration and Assessment of Prior Consideration of the CDM, EB62, Annex 13, Version 04	
/9/	UNFCCC, Glossary of CDM terms, Version 05	
/10/	/10.1/	Project Design Document (PDD), Version 01 at GSP, 12 th July 2010
	/10.2/	Project Design Document (PDD), Version 03, 18 th April 2011
/11/	IRR spreadsheet of Longxin Project	
/12/	EF and ER spreadsheet of Longxin Project	
/13/	CRWQC, Business Registration Certificate (No. 370283410001468), Company Establishment: 10 th March 2011.	
/14/	CRWQC, Company Statute Article, December 2010	
/15/	Shandong Provincial Industry Design Institute Co., Ltd, Feasibility Study Report of the project, 28 th December 2010	
/16/	Development and Reform Committee of Shandong Province, Approval of the Project, LuFaGaiWaiZi(2011)No.154, 2 nd March 2011. Development and Reform Committee of Shandong Province, Approval of the Project Transfer to CRWQC, 18 th November 2011.	

/17/	CRWQC, The Training Plan of Qingdao Longxin Wind Farm, 14 th Dec. 2011
/18/	CRWQC and Easy Carbon Consultancy Co., Ltd., CDM Consultancy Contract, 25 th April 2011
/19/	Qingdao Branch, Communication Bank Company Limited, Loan Contract. No. 2012 PingDuGuDai001, 30 th March 2012
/20/	CRWQC, Management Board Meeting, Decision on implementing the investment of the project with the CDM application. 15 th March 2011
/21/	CRWQC, Project Contract List, December 2011
/22/	Changchun Guodian Construction Supervision Co., Ltd. Order to Commencement of the project, 26 th May 2011
/23/	CRWQC, ODA Declaration Document, 1 st February 2012
/24/	CRWQC and Sichuan Transmission Line Construction Co., Ltd. 35kV Transmission Line Construction Contract. 4 th Aug. 2011.
/25/	CRWQC and Jiansu Zhongzhou Industrial Trading Co., Ltd. Purchase Contract of Tower Rack, CRPW-PD-2011-001, 1 st April 2011
/26/	CRWQC and Shandong Xing'an Iron Tower Co., Ltd. Tower Purchase Contract, No. CRNE-HSDPD-2011-088, 10 th August 2011
/27/	CRWQC and Vestas Wind Technology (China) Co., Ltd. Purchase Contract of Turbine Equipment and related Service. No. CRNE-HSDPD-2011-021, 30 th March 2011
/28/	CRWQC and Jinan Transformer Group Co., Ltd. Purchase Contract of 110kV Transformer, No CRNE-HSDPD-2011-013. 10 th April 2011.
/29/	Qingdao City Environmental Protection Science Research Institute, EIA Report Table of project, No. HB 102068, September 2010
/30/	Qingdao City Environmental Protection Bureau, EIA approval. QingHuanShen(2010)No.209, 12 th Oct. 2010
/31/	CRWQC, Project Construction Schedule, January 2011
/32/	UNFCCC, PDD version 1 (12 th July 2010) was made publicly available on CDM's website: From 10 th September 2011 to 9 th October 2011 (http://cdm.unfccc.int/Projects/Validation/DB/WEYY5DMTUSY5J5GLVWH0NVQ7AW88KL/view.html)
/33/	CRWQC and China 17 th Metallurgical Group Corporation, Construction Contract of the Project (incl. road, basement, platform, wind turbine installation, Thunder Protection and Grounding System, Substation) CRNE-HSDPD-2011-027, 1 st May 2011.
/34/	CRWQC and Land Resource Bureau of Pingdu City, Land compensation agreement
/35/	China National Development and Committee, Notification on the policies of grid-tariff of wind power in China. Document No.: FaGaiJiaGe(2009)No. 1906. 20 th July 2009
/36/	CRWQC and TÜV Rheinland Japan Ltd. (TUVR), CDM validation service contract, 4 th July 2011
/37/	Notification to UNFCCC, UNFCCC received the notification on 17 th October 2011
/38/	NDRC, 《Design Budgetary Estimate Method and Calculation Standard for Wind Farm Project 》, FD001, 2007

/39/	The Layout and geographic Coordinates of the 25 turbines of the Project,
/40/	Minutes meeting of stakeholder consultation in Pingdu City, 49 questionnaires were attached. March 2011
/41/	State Power Cooperation of China, Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects, 2003
/42/	Verification Regulation of Electrical Energy Meters with Electronics (JJG596-1999)
/43/	Revised 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Table 1.4, Page 1.24, Chapter 1, Volume 2
/44/	Technology & Management Regulations for Power Metering Device (DL/T448-2000).
/45/	Shandong Electricity (Group) Co. Ltd. The Approval of Grid-in Connection Program of Longxin Wind Farm. LuDianJiTuanFaZhan(2011)No.65, 30 th January 2011
/46/	CRWQC, Operation Procedures of Longxin Project, 2011
/47/	CRWQC, Maintenance Procedures of Longxin Project, 2011
/48/	CRWQC, CDM training plan and training records for the project. 1 st and 16 th May 2011
/49/	Shandong Provincial Industry Design Institute Co., Ltd, The Explanation of the Electricity Generation Quantity of Qingdao Longxin Wind Farm Project, 2 nd February 2012
/50/	Shi-Pengfei, The Statistical of Installation Capacity of Wind Farm in China in 2007, 2008, 2009.
/51/	China NDRC, the qualification of "Shandong Provincial Industry Design Institute Co., Ltd", No. GongZiYi11820070043, 10 th Dec. 2007
/52/	Office of National Coordination Committee on Climate Change, Baseline Emission Factor Calculation Result of China Grids in 2010, 20 th December 2010 (http://cdm.ccchina.gov.cn/WebSite/CDM/UpFile/File2552.pdf)
/53/	China Electric Power Yearbook Editorial Board, China Electric Power Yearbook 2006 to 2009
/54/	Department of Industry and Transport Statistics, National Bureau of Statistics & Energy Bureau, National Development and Reform Commission, People's Republic of China, China Energy Statistics Yearbook 2006 to 2009
/55/	China State Council, <Notice on strictly prohibiting the installation of thermal power units with capacity of 135MW or below> (Ref. No.: GuoBanFaMingDian [2002](6)), 15 th April 2002
/56/	CRWQC, Notify to China DNA, 19 th September 2011
/57/	GB/T18710-2002 <Wind Farm Wind Resource Assessment Method>, 2002.
/58/	http://www.pbc.gov.cn/ , the monetary policy of THE PEOPLE'S BANK OF CHINA, from 21 st Oct. 2010.
/59/	"Interim regulations of Educational Surtax" and "Interim regulations of Local Educational Surtax in Shandong Province", 4% of VAT is indicated.

	http://www.gov.cn/gongbao/content/2005/content_91662.htm http://www.qdedu.gov.cn/qdedu/6dad12c-bed5-462d-9c49-bd07e8082212/29dab101-0fe1-443d-89ca-591172f9aed7.html
/60/	State Administration of Taxation, Corporation Income Tax Law (Ref.: President of the People's Republic of China Order [2007] no.63), (http://202.108.90.146/guoshui/action/GetArticleView1.do?id=43127&flag=1)
/61/	National Bureau of Statistics of China, China Statistic Year Book, Year 2009, Chapter 9-11, chapter 10-5 http://www.stats.gov.cn/tjsj/2010/indexch.htm
/62/	UNFCCC, "Information Note on the Highest Tariffs Applied by the Executive Board in its decisions on Registration of Projects in the People's Republic of China"(version 02).
/63/	http://www.pd-forum.net/files/0be74ab648713381bb267005dc7e208e.pdf , the article shows the project is a demonstration project of local government and also received ODA. July 2010
/64/	http://www.xindianli.com/xinnenyuan/201103/24-259555.html , Statistical of China Wind Power Project Installation Capacity, 2010
/65/	Land Resource Department of Shandong Province, Land Use Pre-Approval, LuGuoTuZi(2010)No.1122, 19 th Oct. 2010
/66/	http://www.cwei.org.cn/7521/1/newscontent.html , Topic: Raw Material Price Booming, Wind Power Enterprises Face Poor Performance, 28 th July 2011,
/67/	CRWQC, Organization Chart of Project, 2011
/68/	China DRC, LoA issued, (English Version) No.3674, February, 2012 http://cdm.ccchina.gov.cn/website/CDM/pdf/Item_new/Item_new7910.pdf
/69/	PP and Thompson@unfccc.int, Cdmregistration@unfccc.int, etc, email communications on the CDM prior considerations of the Project. August to Oct. 2011.
/70/	CRWQC, China DRC, signed MoC, 28 th March 2012.

2.2 Follow-up Interviews with Project Stakeholders

Table 2: The following table identifies the personnel who have been interviewed and/or provided additional information to the presented documentation

Date	Name	Organization	Title
2011-10-10	Mr.Li Chun	DRC of Pingdu City	Section Head
	Mr. Sun Weimin	EPB of Pingdu City	Section Head
	Mr. Gao Tianshan	Cui Jiaying Village	Local resident
	Mr. Shan Xingmin	Shuan Miao Village	Local resident
	Mr. Meng Fanshi	Zhujiia Jing Village	Local resident
	Mr. Wei Donglin	China Resources Wind Power	Project manager (representative of

		(Qingdao) Co., Ltd.	the Project Owner)
	Mr. Mao Liguao		Project manager
	Mr. Luo Yifeng		CDM manager
2011-10-11	Ms. Yang Fang	Easy Carbon Consultancy Co., Ltd.	Project manager
	Ms. Hou Chunfeng		Project manager

2.3 Resolution of Outstanding Issues

The objective of this phase of the validation is to resolve any outstanding issues which need be clarified prior to TÜV Rheinland's conclusion on the project design. In order to ensure transparency a validation protocol is customised for the project. The protocol shows in transparent manner criteria (requirements), means of verification and the results from validating the identified criteria. The validation protocol serves the following purposes:

- It organises, details and clarifies the requirements a CDM project is expected to meet;
- It ensures a transparent validation process where the validator will document how a particular requirement has been validated and the result of the validation.

The validation protocol consists of two tables. The different columns in these tables are described in the figure below. The completed validation protocol for this project is enclosed in Appendix A to this report.

Findings established during the validation can either be seen as a non-fulfilment of CDM criteria or where a risk to the fulfilment of project objectives is identified. Corrective action requests (CAR) are issued, where:

- mistakes have been made with a direct influence on project results;
 - CDM and/or methodology specific requirements have not been met; or
 - there is a risk that the project would not be accepted as a CDM project or that emission reductions will not be certified.
- A request for clarification (CL) may be used where additional information is needed to fully clarify an issue.

Herewith, the Validation Team summarizes *major* changes between webhosted PDD and final version of PDD for submission as follows:

	Webhosted PDD	Correction to webhosted PDD in the final PDD submission for registration with DOE acceptance.
Methodologies	ACM0002 Version 12.1.0	ACM0002 Version 12.2.0

CER calculations	92,183 tCO ₂ e/yr	No change
Additionality	The additionality was demonstrated by Benchmark Analysis: the project IRR is 5.63% without considering the CDM revenue, lower than the benchmark of 8%, and the common practice analysis was correctly done.	No change
Monitoring	The necessary training and the QA/QC procedure applied in the GSP-PDD are not clear enough.	The necessary training and the QA/QC procedure applied are recited in the monitoring plan.
Please refer to Appendix A of this report for details of each change between webhosted PDD and the final PDD for submission. The Validation Team has carried out the validation process based on the Webhosted PDD and raised CARs/CLs against the project by issuing the validation protocol. With the updated information and corrections done on final PDD, the PP has addressed all the CARs /CLs that were raised by the Validation Team. It is concluded that the Validation Team has reviewed the project in line with the VVM (version 01.2) and all the evidence, corrections, justifications and updating done on the final PDD with respect to CARs /CLs raised are accepted and closed by the Validation Team, issuing the positive validation opinion for project registration. No FAR was issued to the DOE verification team to check the implementation and operational completeness during the first verification.		

Validation Protocol Table 1: Requirement checklist				
Checklist Question	Reference	Means of verification (MoV)	Comment	Draft and/or Final Conclusion
<i>The various requirements in Table 1 are linked to checklist questions the project should meet. The checklist is organised in different sections, following the logic of the small-scale PDD template, version 03 - in effect as of: 22nd December 2006. Each section is then further subdivided.</i>	<i>Gives reference to documents where the answer to the checklist question or item is found.</i>	<i>Explains how conformance with the checklist question is investigated. Examples of means of verification are document review (DR) or interview (I). N/A means not applicable.</i>	<i>The section is used to elaborate and discuss the checklist question and/or the conformance to the question. It is further used to explain the conclusions reached.</i>	<i>This is either acceptable based on evidence provided (OK), or a corrective action request (CAR) due to non-compliance with the checklist question (See below). A request for clarification (CL) is used when the validation team has identified a need for further clarification.</i>

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Validation Protocol Table 2: Resolution of Corrective Action and Clarification Requests			
Draft report clarifications and corrective action requests	Ref. to checklist question in table 2	Summary of project owner response	Validation conclusion
<i>If the conclusions from the draft Validation are either a CAR or a CL, these should be listed in this section.</i>	<i>Reference to the checklist question number in Table 1 where the CAR or CL is explained.</i>	<i>The responses given by the project participants during the communications with the validation team should be summarised in this section.</i>	<i>This section should summarise the validation team's responses and final conclusions. The conclusions should also be included in Table 2, under "Final Conclusion".</i>

• Figure 1. Validation protocol tables

2.4 Internal Quality Control

The validation report including the validation findings underwent a technical review before requesting registration of the project activity. The technical review was performed by a technical reviewer qualified in accordance with TÜV Rheinland's qualification scheme for CDM validation and verification.

2.5 Validation Team

Validation Team			Type of Involvement						
Full name	Affiliation TÜV Rheinland	Appointed for Sectoral Scopes (Technical Areas)	Supervising the work	Desk review	Site Visit + Interview	Report and protocol Writing	Technicia Expert Input	Reporting Support	Technical Reviewer
Mr. Minglong Huang	China	1.2, 5.1, 11.1, 12.1	X	X	X	X			
Mr. Lixin Li	China	1.1, 1.2, 3.1							X

3 VALIDATION FINDINGS

The findings of the validation are stated in the following sections. The validation criteria (requirements), the means of verification and the results from validating the identified criteria are documented in more detail in the validation protocol in Appendix A.

The final validation findings relate to the project design as documented and described in the revised and resubmitted project design documentation.

3.1 Approval and Participation

Validation Opinion: The project is a unilateral CDM project which was conducted by project participant: China Resources Wind Power (Qingdao) Co., Ltd. (CRWQC) from host party, P.R. China.

China Resources Wind Power (Qingdao) Co., Ltd. is a local investment based entity, which was established on 10th March 2011 /13/. The host party, P.R. China meets all relevant participation requirements in CDM. The Letter of Approval (LoA) /68/ of voluntary participation, including confirmation by China's DNA – National Development & Reform Commission (NDRC), that the project assists China in achieving sustainable development, has been received. Hence, voluntary participation of CRWQC is confirmed.

The project proponent planned to raise funding by internal accrual and the loan from the Bank, validation team verified the loan contract /19/ signed with Qingdao Branch, China Communication Bank Company Limited and confirmed it is valid. This is also confirmed by interviewees /vi/ /vii/ during OSV. It is also confirmed from a statement that non ODA capital was involved in the project by the project owner/23/. During on-site visit (OSV), validation team interviewed local government officials /i/ to /iii/ and project owner /vi/ to

/vii/, did not reveal any information to indicate that the project can be seen as a diversion of ODA funding towards China.

Table 4: Summary of project participants and parties involved.

Project participants	China Resources Wind Power (Qingdao) Co., Ltd.
Parties involved	P.R. China (host)
APPROVAL	
LoA received	Yes
Date of LoA	Feb. 2012
Reference to document	No. 3674
LoA received from	PP
Validation of authenticity	Yes. Crosschecked from LoA approval list in P.R. China ¹
Validity of LoA	Yes
PARTICIPATION	
Party is party to Kyoto Protocol	Yes Confirmed from UNFCCC web (http://unfccc.int/files/kyoto_protocol/status_of_ratification/application/pdf/kp_ratification.pdf)
Voluntary participation	Yes. Statement 3 of LoA from China DNA
Diversion of official development aid towards host country	N/A
Project contribution to SD	Yes. Statement 2 of LoA from China DNA

The MoC /70/ was also received and confirmed to be valid.

3.2 Project Design Document

Validation Opinion: The validation team validated that the provided Project Design Document (PDD) version 03 which is based on the currently valid PDD template /2/ and is completed in accordance with the applicable guidance document /3/.

3.3 Project Description

Validation Opinion: The “Qingdao Longxin Wind Power Project Phase I” is a wind farm project which is located in Liyuan Town and Dianzi Town of Pingdu City, Shandong Province, P.R. China with a range geographical coordinates of 119.8827° to 119.9503° east longitude, and 36.8157° to 37.8915° north latitude for the project, the central GPS coordinate is 119.91125°E, 36.8568°N. the coordinates of each turbines were listed in a project location sheet/39/, which were counterchecked by the validation

¹ Approval by NDRC, http://cdm.ccchina.gov.cn/website/CDM/pdf/Item_new/Item_new7910.pdf

team via public online source (i.e. Google Earth, etc.) and document review /15/ /29/ and confirmed correct. The nearest Liyuan Town is about 10 kilometers from the project site. According to FSR/15/, the average wind speeds are 5.88m/s at 10m height and 6.69m/s at 70m wind measurement tower respectively within the project site while the annual average wind density is 228.05W/m² at 10m height and 319.57W/m² at 70m height wind measurement tower respectively/15/.

The project consists of 24 sets of 2,000KW wind turbines and generators (Models:V90-2.0MW) and 1 set of 1,800kW wind turbine and generator (Model:V90-1.8MW), with a total capacity of 49.8MW. It was found that wind turbine and generator's models are consistent between PDD /10/ and equipment purchase contract /27/. The chosen turbines have vane diameters of 90m and tower hub height of 80m/15/. The major equipments of the project are supplied by Vestas Wind Technology (China) Ltd /25/ without foreign technology transfer. According to the PDD, the net annual electricity supply from the project activity is 99,026MWh. The electricity generated from the project activity is expected to substitute the equivalent amount of power supplied from North China Power Grid (NCPG) which is mainly composed of fossil fuel-fired power plants. The expected GHG emission reductions of the project activity are 92,183 tCO₂e annually during the first 7-years. There is no leakage considered in this project since this is a GHG zero-emission project as per the applied methodology.

The Feasibility Study Report (FSR) was prepared by Shandong Provincial Industry Design Institute Co., Ltd (here after as SPIDL) in Dec. 2010 /15/. SPIDL was granted by the government for the professional accreditations (Ref No.: Class B certificates no. 11820070043) /51/ in the fields of general engineering survey and consultation in 10th Dec.2007. The Project was approved by Development and Reform Committee of Shandong Province (DRC) on 2nd March 2011/16/. The annual utilization hour for net power supply is 1,988 hours and the plant load factor is 0.227 as indicated in the FSR. The validation team confirms that the determination of PLF is in accordance with "Guidelines for the reporting and validation of plant load factors" (Version 01) since FSR writer, i.e. SPIDL is the qualified and independent party, fulfilling option (b) of the guideline.

According to "Grid-in Connection Program"/45/ which was approved by Shandong Electricity (Group) Co. Ltd, the project will share the on-site booster station (35kV/110kV) then connected to 220kV Tangtian Transformer Substation to Shandong Power Grid which is a sub-power grid of NCPG with the power generation system of the project activity. The project boundary based on the electricity connection system can be confirmed.

During the OSV with project owner /vi/ and CDM consultant /ix/ /x/, it was understood that the project owner would be responsible for organizing the necessary technical training for the daily operation, maintenance and monitoring of wind farm. CDM monitoring training plan was scheduled to be implemented just after the project successfully registered in the UNFCCC and CDM monitoring procedure in PDD have been described clearly. Project-specific operation manual and maintenance procedure /46/ /47/, and the general training plan /17/ of the project was provided to the validation

team, the training for operating staff would be carried out in later stage before completion of project construction. Regarding the project-specific training for CDM implementation, it will be carried out by the CDM consultancy company, i.e. Easy Carbon Consultancy Co., Ltd.

As observed during OSV, the construction of the project has been started, the project was scheduled to put into operation in December of 2011 according to the project schedule /31/.

According to the PDD, a renewable crediting period of 7 years is selected. The starting date of project activity, which was the date of signing the Turbine Purchasing Contract on 30th March 2011. According to the definition in Glossary of CDM terms, by review the Turbine Purchasing Contract /27/, Tower Purchasing Contract /26/ and the construction commencement order/22/, and construction contract /33/, it is found that the earliest date of these actions is 30th March 2011, i.e. the signing date of Turbine Purchasing Contract. The starting date of the first crediting period is indicated as Jul.20 2012 or the date of registration, whichever is later. The expected operational lifetime of the project activity is 20 years. Further discussion will be covered in Section 3.5.1.

Table 5: Critical project description milestones

Starting date of project	Expected project operational lifetime	Crediting period
30 th March 2011 (the signing date of Turbine Purchasing Contract)	20 years	Jul.20 2012 to Jul.19 2018 Starting date: Jul.20 2012 or the date of registration, whichever is later.

In summary, according to clause 58 of the VVM /1/, by means of document review, searching information, observation of OSV, stakeholders' interviews, it is found that PP can clearly demonstrate the precise nature of the project activity and the technical aspects of its implementation.

3.4 Baseline and Monitoring Methodology

3.4.1 Applicability of the selected methodology to the project activity

Validation Opinion: The project activity involves electricity generation for a grid through wind sources where the output capacity is 49.8MW. It correctly applies the ACM0002/Version 12.2.0 "Consolidated baseline methodology for grid-connected electricity generation from renewable sources", which is available to be applied since EB58. The approved methodology is also in connection with methodological tools, "Tool for the demonstration and assessment of additionality" Version 05.2.1" and "Tool to calculate the emission factor for an electricity system/ Version 02.2.0".

Applicability criteria for the baseline and monitoring methodology are assessed by the validation team by means of document review and interviews. The proposed project fully met the criteria set out in the methodology as described below:

- The project activity consists of 25 sets of wind farm turbine project which generates electricity by utilizing the renewable wind resource, which supplies electricity to and displaces electricity from the electricity distribution system, i.e. North China Power Grid (NCPG). NCPG mainly comprises fossil fuel-fired power plants which have been supplied by more than one fossil fuel-fired generating unit;
- The project activity does not involve “switching from fossil fuels to renewable energy sources”;
- The geographic and system boundaries for the NCPG can be clearly identified and information on the characteristics of the grid is available.

Based on the above, it is the validation team’s opinion that the methodology ACM0002/Version 12.2.0 is correctly applied to the project activity.

According to the physical inspection of project site, the validation team did not observe any fossil fuel fired equipment at the project location for power supply to the project, the project would not contain any fossil fuel fired generator during the whole project period. Hence, the validation team realized no other major sources of emission reductions which are not addressed by the applied methodology, i.e. ACM0002/Version 12.2.0 is deemed to contribute more than 1% of the overall expected average annual emissions reductions. Please refer to Section 3.4.4 for the detailed discussion.

3.4.2 Project Boundary

Validation Opinion: The project boundary includes the project power plant and all power plants connected physically to the electricity system that the CDM project power plant is connected to. Here it is defined as the NCPG including the Beijing City, Tianjin City, Hebei Province, Shandong Province Shanxi Province and Inner Mongolia Autonomous Region Power Grid on the basis of information announced by NDRC /52/ on 20th Dec. 2010. There are no significant transmission constraints between the power plants of the NCPG.

The validation team has validated the relevant documents of the project, i.e. FSR/15/, Approval of grid-in connection /45/, and on-site observation, and then confirmed that the selected project boundary is correct. The project activity is to construct a new wind farm with 49.8MW total installed capacity. The wind farm can mainly be classified into three parts, (i) 25 sets of wind turbine with associated power generators, (ii) main control house, and (iii) power cable system inside wind farm to power grid. The locations of all parts of the wind farm are in Liyuan Town and Dianzi Town in Pingdu City, Shandong Province, P. R. China. The project will only generate electricity by utilizing wind resources and supply on grid, i.e. NCPG via firstly stepping up to 110kV in project site, next to local power grid, then to Shandong Province Power Grid and finally to NCPG. All project activities are bounded in NCPG. It is also noted that there is no electricity exported from NCPG to other Grid, and there is net electricity imports from Central

China Power Grid (CCPG) to NCPG/52/. The baseline emission and the project emissions are shown in the following table:

Table 6: Presentation and transparent justification on the system boundary of the project

	GHGs involved	Description
Baseline emissions	CO ₂	Major emission source, which is emitted from the electricity generation by fossil fuel-fired power plants connected to NCPG
Project emissions	--	No use of supplementary fossil fuel is observed during OSV. Project emission is considered as zero as per methodology ACM0002/Version 12.2.0
Leakage	--	No leakage emissions are considered as per ACM0002/Version 12.2.0.

3.4.3 Baseline Identification

Validation Opinion: The project activity is the installation of a new grid-connected renewable power plant. According to the applied methodology ACM0002/Version 12.2.0, the baseline scenario is prescribed as: "Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin(CM) calculations described in the "Tool to calculate the emission factor for an electricity system."

The validation team has checked the following in accordance with the latest version of CDM Validation and Verification Manual /1/, and the results are tabulated as follows. The details can be referred to Appendix A.

Table 7: Summary on the baseline identification

1. The approved baseline methodology applicable to the project - explicit criteria - implicit criteria (e.g. available scenarios, applicability of formulas for BE/PE/LE calculations)	☒ Yes ☐ No	As per Section II of the ACM0002/Version 12.2.0, the baseline scenario is prescribed. Please refer Section 3.4.1 for details
2. PDD includes all assumptions and data used by project participants	☒ Yes ☐ No	As per section II of the ACM0002/Version 12.2, the assumptions are excluded since it is a new wind power project.
3. All the references and documents used are relevant for establishing the baseline scenario	☒ Yes ☐ No	As per section II of the ACM0002/Version 12.2.0, the baseline scenario is prescribed. Yes, all the reference and documents used are relevant for

		establishing the baseline scenario, i.e., the relevant regulations and the publications in the raised industry sector.
4. All the references and documents used are correctly quoted and conservatively interpreted in the PDD	☒ Yes ☐ No	As per section II of the ACM0002/Version 12.2.0, the baseline scenario is prescribed. Yes, the validation team verified all the references and documents through crosschecking and found they are correctly cited and conservatively interpreted in the PDD.
5. All relevant policies / regulations considered are listed in the PDD	☒ Yes ☐ No	As per section II of the ACM0002/Version 12.2.0, the baseline scenario is prescribed. All the relevant policies / regulation have been considered and listed in the PDD. It was verified by site validation and desk review for the project.
6. Identified potential baseline scenarios reasonably represent what would/could occur in the absence of the proposed project activity:	☒ Yes ☐ No	As per section II of the ACM0002/Version 12.2.0, the baseline scenario is prescribed. This project is a new grid-connect wind farm project which was confirmed from FSR and through site visit.
7. The baseline scenario selection is appropriate and determined according to the methodology	☒ Yes ☐ No	As per section II of the ACM0002/Version 12.2.0, the baseline scenario is prescribed.
8. The approved methodology used is applicable to the identified baseline scenario	☒ Yes ☐ No	As per section II of the ACM0002/Version 12.2.0, the baseline scenario is prescribed.

3.4.4 GHG Emission Reductions

Validation Opinion: The GHG emissions calculations are transparently documented and appropriate assumptions regarding expected amount of electricity generated have been used to forecast emission reductions.

According to the selected methodology ACM0002/Version 12.2.0 for grid-connected renewable electricity generation and the “Tool to calculate the emission factor for an electricity system”/Version 02, the emission reduction (ER_y) by the project during the crediting period is the difference between the baseline emissions (BE_y), project emissions (PE_y) and emissions arising from leakage (L_y), which is expressed as follows:

$$ER_y = BE_y - PE_y$$

Regarding the calculation of project emissions, since the project activity utilizes renewable wind source for electricity generation where no auxiliary fuels shall be used and it is verified by the validation team during OSV. The project emission is zero as per ACM0002.

According to the ACM0002/Version 12.2.0, No leakage emissions are considered.

While the project emission & leakage are zero, the baseline emission is equal to the emission reductions due to the project activity. According to ACM0002/Version 12.2.0, the baseline emissions are demonstrated in Section 6 of PDD and are calculated as follows:

$$BE_y = EG_{PJ,y} * EF_{grid,CM,y}$$

Where:

- BE_y = Baseline emissions in year y (tCO₂e/yr);
- $EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr);
- $EF_{grid,CM,y}$ = Combined margin CO₂ emissions factor for grid connected power generation in year y.

The project involves the installation of a new wind power project, which is defined as a greenfield renewable energy power plant,

$$EG_{PJ,y} = EG_{facility,y}$$

Where:

- $EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr);
- $EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr).

Therefore, $EG_{facility,y}$ is equal to the net annual power supply to grid by the project, which is 99,026MWh.

$$EG_{PJ,y} = EG_{facility,y} = 99,026\text{MWh.}$$

The baseline emission factor for the project, using the combined margin (CM) approach, is fixed ex-ante during the first crediting period. According to ACM0002/Version 12.2.0 and the linked tool, "Tool to calculate the emission factor for an electricity system/Version 02.2.0", the default weights for the proposed project are 75% for OM and 25% for BM for the first crediting period (owing to their intermittent and non-

dispatchable nature). For the calculation of OM emission factor, simple OM emission factor calculation method is chosen because low cost must run projects constitute less than 50% of the total grid generation and data is not available for applying the dispatch data analysis OM and simple adjusted OM. The average emission factor for the grid for each fuel type is calculated ex-ante based on a 3-year full generation-weighted average of the most recent statistics available (data available from 2006, 2007 and 2008 derived from China Electric Power Yearbook 2007-2009 and China Energy Statistics Yearbook 2006, 2007 and 2008 respectively at the time of PDD submission²). Because plant specific fuel consumption and electricity generation data is not publicly available in China for BM calculation, a deviation of the baseline methodology of AM0005 (later replaced by ACM0002) approved by the EB (as detailed in the PDD) is adopted for using relevant emission data recently published by the DNA of P.R. China on 20th Dec. 2010 /52/, incorporating the latest data from the China Electric Power Yearbook 2009 /53/ and China Energy Statistics Yearbook 2009 /54/, in which the data of installed capacities of power plants in year 2006, 2007 and 2008 were used for the recent increment of 20% of installed capacities in the NCPG.

The validation team has checked the calculation in Annex 3 of PDD /10.2/ and confirmed that the calculation of CM is based on the requirement in “Tool to calculate the emission factor for an electricity system/Version 02.2.0”. The validation team has verified the parameters applied in the baseline information according to the data source, i.e. China Electric Power Yearbook and China Energy Statistical Yearbook 2007-2009 /50/ /51/, 2006 IPCC Guideline for National Greenhouse Gas Inventories, Volume 2 /43/. The applied parameters are consistent within the reference materials. The simple OM emission factor was calculated as 0.9914tCO₂e/MWh and the BM emission factor as 0.7459tCO₂e/MWh, resulting in the CM as 0.9309 tCO₂e/MWh. Therefore, the emission reductions due to the project activity were estimated ex-ante to be 92,183 tCO₂e per year in the PDD and calculated as follows:

$$ER_y = BE_y = EG_{PJ, y} * EF_{grid, CM, y} = 99,026MWh * 0.9309 tCO_2e/MWh = 92,183 tCO_2e.$$

The ex-ante estimation of emission reductions is based on the most recent data available at the time of submission of the PDD to the DOE for validation (i.e. GSP, 10th September to 9th October 2011), for the net electricity generation and the relevant baseline emissions and project emissions of the project, which is reasonably and transparently carried out.

Table 8: Summary on the GHG emission reduction of the project

1. All assumptions made for estimating GHG are listed in the PDD	☒ Yes ☐ No	As per PDD Section B.6. Emission Reduction
2. All data used by project participants are listed in the PDD	☒ Yes ☐ No	As per PDD Annex 3 Baseline Information

² PDD submission for validation(i.e. GSP) was on 9th Feb. 2010, version 03, the GSP period is from 9th Feb. 2010 to 10th March 2010.

3. Their references and sources are also listed in the PDD	☒ Yes ☐ No	As per PDD Annex 3 Baseline Information & Section B.6
4. Formulas, parameters, values are complete, accurate, transparent and conservative	☒ Yes ☐ No	As per PDD Annex 3 Baseline Information
5. All the references and documents used are correctly quoted and conservatively interpreted in the PDD	☒ Yes ☐ No	As per PDD Section B.6. Emission Reduction and Annex 3 Baseline Information
6. Methodology has been applied correctly to calculate project emissions, baseline emissions, leakage emissions and emission reductions	☒ Yes ☐ No	As per ACM0002/Version 12.2.0.
7. All the emissions of baseline emissions can be replicated using information provided in the PDD	☒ Yes ☐ No	As per PDD Section B.6. Emission Reduction and Annex 3 Baseline Information

3.5 Additionality

As per ACM0002/Version 12.2.0, the project proponent has applied the “Tool for the demonstration and assessment of additionality” Version 05.2.1 approved by the EB to demonstrate the additionality of the project.

3.5.1 CDM consideration

Validation Opinion: As per EB 62 Annex 13, for project activities with a starting date on or after 02 August 2008, the project participant must inform a Host Party DNA and the UNFCCC secretariat in writing of the commencement of the project activity and of their intention to seek CDM status. Such notification must be made within six months of the project activity start date and shall contain the precise geographical location and a brief description of the proposed project activity, using the standardized form F-CDM-Prior Consideration”.

The PP abides by the above requirement and the related CDM development progress of the project is shown in the table below:

Table 9: The CDM development milestones of the project

Date	History	
	Project related	CDM related
September 2010	Completion of EIAR (Ref./29/)	

Date	History	
	Project related	CDM related
December 2010	Completion of Feasibility Study Report (FSR) (Ref./15/)	
March 2011	Project Approval (Ref./16/)	
30 th March 2011	Signing of Turbine Purchasing Contract(project starting date) (Ref. /27/)	
25 th April 2011		Signing of the CDM consultancy agreement (Ref./18/)
19 th May 2011	Signing of Construction Contract for the Project(Ref. /33/)	
4 th July 2011		Signing of CDM Validation Service Agreement (Ref./36/)
August to Oct. 2011		A few emails between PP and Thompson@unfccc.int, Cdmregistration@unfccc.int, etc (Ref./69/)
10 th September 2011		PDD was published in UNFCCC website (Ref./32/)
19 th September 2011		Notifying China DNA (NDRC) with the CDM notification form(Ref./38/)(Ref./56/)
17 th October 2011		Notifying UNFCCC with the CDM notification form (Ref./37/)

From the above project history of CDM development, it is verified by the validation team that the project starting date is Signing of the Turbine Purchasing Contract on 30th March 2011 which is the earliest real action of the project (i.e. the next real action is the signing of Construction Contract, on 19th May 2011), then, PP signed the validation service contract with DOE in 4th July 2011/36/, and PP published the PDD in UNFCCC website on 10th September 2011/32/, at the same time, PP went to do the Notification to the China DNA and the UNFCCC on 19th September 2011/56/ and 17th October 2011/37/ respectively (a few email communications between the PP and UNFCCC /69/ were attached to explain the cause of “a few days delay”). This could comply with the Para. 2 of EB62 Annex 13, “When validating a project activity with a start date on or after 2 August 2008, DOEs shall ensure by means of confirmation from the UNFCCC

secretariat that such a notification had been provided within 6 months of the project starting date”

In conclusion, according to the project’s milestones described above and the evidences from the CRWQC, the validation team considers that CDM revenue has been considered prior to the project implementation.

3.5.2 Alternatives

Validation Opinion: According to VVM (i.e. clause 115), the alternatives of the proposed project is not needed to be identified and analyzed as the baseline scenario has been prescribed in the ACM0002 (Version 12.2.0) applied.

3.5.3 Investment analysis

Validation Opinion: According to “Tool for the demonstration and assessment of additionality”, Version 05.2.1, an investment analysis was carried out through applying benchmark analysis method, which is mainly based on the comparison between project IRR and benchmark rate. The simple cost analysis and investment comparison analysis are excluded because: 1) the proposed project will have the income not only from CDM revenues but also the sale of electricity, the simple cost analysis method (option I) is not appropriate. 2) The alternative of the proposed project is the NCPG rather than new investment project, they are not similar investment project. The validation team confirms the choice is valid.

The project participant applied the economic analysis reference for the China power sector, i.e. “Interim Rules on Economic Assessment of Electric Engineering Retrofit Projects” issued by the State Power Corporation of China in 2002. The financial benchmark IRR of the Chinese electricity industry from the benchmark reference is 8% (post tax) on total investment (i.e. project IRR = 8%). It applies to the P.R. China power companies for power generation and transmission projects as stated in section 1.15. According to section 1.11, the economic benchmark is regulated as 8% on total investment, and this is deemed an appropriate benchmark reference for the project activity. In addition, the benchmark reference document has been commonly adopted for financial evaluation of power projects for the approved renewable power CDM projects in China.

The following is an excerpt extracted from the FSR /15/ and presents the guideline documents referenced in the investment analysis which demonstrates that the inputs values adopted for the investment analysis are in accordance with appropriate guidelines and best available market information:

1. *The China State Power Corporation: 《Interim Rules on Economic Assessments of Electric Engineering Retrofit Projects》 /41/*
2. *NDRC, 《Design Budgetary Estimate Method and Calculation Standard for Wind Farm Project 》, FD001, 2007 /38/*
3. *GB/T 18710-2002 《Methodology of Wind Energy Resources Assessment for Wind Farm 》 /57/*

According to the PDD, the benchmark analysis shows that, without the revenue from CDM, the project IRR would be 5.63%, which is below the benchmark IRR of 8%. Hence, subject to further clarification on the applicability of the selected benchmark, the project cannot be considered as financially attractive in the absence of CDM benefits.

The project IRR would be improved in consideration of CERs revenue, where project IRR of 8.36% shall be expected under the assumed CER price of EUR10.5 / tCO₂e.

According to EB62 meeting (decision 13), the validation team validated the provided FSR /15/ found that the completion date of FSR is Dec. 2010. and the top management made the decision to launch the project on 15th March 2011, Therefore, validation team can justify the period of time between the finalization of the FSR and the investment decision is sufficiently short (approximately 3 months); Hence, the input values of the project activities are unlikely materially changed.

In accordance with the “Guidance on the Assessment of Investment Analysis (Version 04)” EB62, the validation team has independently checked the following critical project and financial input values for the IRR calculation as Investment Analysis basis. The critical parameters showed in table 2 and table 3, section B.5 of PDD are tabulated as follows:

Table 10: Validation of input value of project finance

Parameters	Value appeared in PDD	Data source validated	Validation Opinions															
Installed Capacity (MW)	49.8	FSR /15/ Turbine Purchasing contract /27/	24 units of 2,000kW wind turbines and associated generators and 1 unit of 1,800kW wind turbine and associated generator: Total power = 2,000KW × 24+1,800x1 = 49.8MW The validation team has checked the equipment purchase contract /27/ and found numbers of unit and unit capacities are correct.															
Total static Investment (CNY in Million)	528.86	FSR /15/ Fiancial Spreadsheet /11/ Project contract list /21/	According the FSR /15/ (i.e. Chapter 13.1), total static investment is CNY528.86million which is consistent with the values indicated in the PDD and financial spreadsheet, The validation team compared the estimated value in the FSR and the actual cost in the main contracts: <table><tr><th>Item</th><th>Value in FSR (10,000 CNY)</th><th>Actual costs by Contracts (10,000 CNY)</th></tr><tr><td>(i)equipment and installation</td><td>44,420</td><td>39,012</td></tr><tr><td>(ii)construction works</td><td>3,535</td><td>5,438</td></tr><tr><td>(i)+(ii)</td><td>51,261</td><td>44,450</td></tr><tr><td>Percentage to the total static investment</td><td></td><td>84.05%</td></tr></table> By referring to sum of actual costs on the contract list /21/ i.e. Wind Turbines’ Purchase Contract /27/, Tower Purchasing Agreement, Agreement /26/, Construction Contract of Foundations and Roads tec./33/, etc., the validation team verified	Item	Value in FSR (10,000 CNY)	Actual costs by Contracts (10,000 CNY)	(i)equipment and installation	44,420	39,012	(ii)construction works	3,535	5,438	(i)+(ii)	51,261	44,450	Percentage to the total static investment		84.05%
Item	Value in FSR (10,000 CNY)	Actual costs by Contracts (10,000 CNY)																
(i)equipment and installation	44,420	39,012																
(ii)construction works	3,535	5,438																
(i)+(ii)	51,261	44,450																
Percentage to the total static investment		84.05%																

			totally 22 contracts, and confirmed they are real and can be traceable. From the sum of contracted amount /21/ above, we can see the actual cost accounts for 90.11% of the total static investment. .In addition, refer to table 11, the comparison to the registered 37 CDM projects in Shangdong Province shows that the average of 10,621.75 CNY/kW in Shandong Province, and the unit kW investment for the project is 10,619.68 CNY. It is better in the range from 7,431.19 CNY/kW to 12,913.94 CNY/kW. All these data were checked by validation team and confirmed traceable and reasonable. Hence, the value of the total static investment is confirmed to be valid.
Annual Utilization Hour (hour)	1,988	FSR /15/	The annual utilization hour is shown in FSR (Chapter 3.1.3) that comes from the 1-year wind speed measurement with 1 wind measurement towel on the project site, it is also based on recent 30 years weather data drawn from Pingdu Weather Station (1979~2008, Chapter 2.2.1 of FSR). The FSR indicated the load factor of project is 0.227, which is derived from: $(1,988 \times 49.8 \text{ MW}) / (8,760 \times 49.8) = 0.227$, The validation team confirms that the determination of PLF is in accordance with "Guidelines for the reporting and validation of plant load factors" (Version 01) since FSR writer, i.e. SPIDL is the qualified and independent party, fulfilling option (b) of the guideline. Validation team also compared the similar projects which were registered in UNFCCC in table 11, the range of utilization hours are from 1,646h to 2,305h, the proposed project with a utilization hour of 1,988h is better in the range. The validation team can therefore reasonably believe that the designed annual utilization hour of 1,988 hr/a is a reasonable statistic data obtained from scientific approach, which is unlikely to have an substantial change in the project lifetime.
Annual net power supply (MWh)	99,026	FSR /10/ GB/T18710-2002 /57/	The validation team is able to check against the FSR (Chapter 6) that the annual nominal electricity generation is 139,159.64MWh which is the alternative 4 in FSR. The validation team is able to check against the FSR (i.e. page 68, Chapter 6) that the annual nominal operating hour is 1,988 hours . The estimated annual nominal operation hour is based on 30-year (i.e. year 1979-2008) wind resource data from the nearby meteorological station and actual wind resource data collection (i.e. June 2009 to May 2010) at pre-project stage. All of wind resource data, calculation procedures are in accordance with national standard, ie. GB/T18710-2002 Wind Farm Wind Resource Assessment Method》 /57/ According to the National Standard, the wind turbines information and data of wind resources such as average wind speed, wind density and annual utilization hours in the wind farm were assessed with professional stimulation software. Based on the assessment results, the theoretical annual power generation of the project is estimated as 139,159.64MWh. Annual power supply = annual power generation x power correction coefficient.

			According to the Chapter 6.2 (i.e. the estimation of annual feed-in electricity) of the FSR /15/, the power correction coefficient of the project was composed of several corrective factors including; (i) Control and turbulent flow reduction factor =4%, which falls into the range of 5% as indicated /57/; (ii) Blade contamination loss factor = 4%, which falls into the range of ≤6% as indicated /57/; (iii) Wind Farm Electricity Consumption and Line Losses = 3%, which falls into the range of 3-10% as indicated /57/. (iv) climate impacts reduction = 4%, which falls into the range of 3-7% as indicated /57/; (v) Loss of Error in software calculation= 5%, which falls into the range of 5-10% as indicated /57/; (vi) wind turbine usable rate reduction = 4%, which falls into the range of 5-10% as indicated /57/; (vii) wind turbine power efficiency = 5%, which falls into the range of 5% as indicated /57/; (viii) the End Flow impact reduction rate =3.3%, which falls into the range of 5% as indicated /57/; (ix) Air Density Correction Rate: 100%, this is conservative for the estimation. Overall power correction coefficient $= 1 \times (1-4\%) \times (1-4\%) \times (1-3\%) \times (1-4\%) \times (1-5\%) \times (1-4\%) \times (1-5\%) \times (1-3.3\%)$ $= 1 \times 0.96 \times 0.97 \times 0.95 \times 0.96 \times 0.91 \times 0.95 \times 0.95 \times 0.967 \times 100\%$ $\approx 71.16\%$ Hence, Annual power supply=139,159.64*71.16%=99,026MWh The Feasibility Study Report (FSR) was prepared by Shandong Provincial Industry Design Institute Co., Ltd (SPIDL) in Dec. 2010 /15/. SPIDL was granted by the government for the professional accreditations (Ref No.: Class B certificates no. 11820070043) /51/ in the fields of general engineering survey and consultation in 10th December 2007. The project was approved by Development and Reform Committee of Shandong Province on 2nd March 2011/16/. Therefore, the validation team could convince the calculations of annual power generation from nominal operating hour and net annual power supply from net operation hours are traceable and deemed realistic.
On-grid tariff (including VAT) (CNY/ kWh)	0.61	FSR /15/ Tariff Regulation /35/	China National Development and Committee, Notification on the policies of grid-tariff of wind power in China. Document No.: FaGaiJiaGe(2009)No. 1906. 20th July 2009 shows the tariff in Shandong Province is 0.61 CNY/kWh (incl. VAT). The reference of tariff in PDD is consistent to indicated tariff in FSR (i.e. Chapter 17.2) indicates the tariff for this project is 0.61 CNY/kWh (incl. VAT). The validation team compared the similar projects which were registered in UNFCCC in Shandong Province in table 11 of this report, it shows that all the projects applied the tariff of 0.61 CNY/kWh (incl. VAT). According to the "Information note on the highest tariffs applied by the Executive Board in its decisions on registration of projects in the People's Republic of China" (version

			02). The highest tariff in Shandong Province is 0.61 Yuan/kWh, Therefore, the estimated tariff of the project during the investment decision is clearly demonstrated with support. The validation team considers CNY0.61/kWh tariff (incl. VAT) is appropriate for the investment analysis of the project.								
Annual O&M cost (CNY in Million)	11.84	FSR /15/ Financial spreadsheet /11/	The annual O&M cost indicates in the "economic evaluation reference data" of Chapter 14.1, FSR /15/. The annual operation cost indicates in "total cost table" after Chapter14 of FSR /15/ and financial spreadsheet. The composition of annual O&M cost is clearly shown as: (i) Maintenance fee: <table><tr><td>Total static investment (TSI)</td><td>52,886 CNY</td></tr><tr><td>Fixed Asset</td><td>52,886-6,097=46,789 6,097 is the equipment input tax</td></tr><tr><td>Maintenance fee</td><td>1.5%*Fixed Asset =46,789*1.5%</td></tr><tr><td>Annual value</td><td>7.0183 million CNY</td></tr></table> (ii) Salary and welfare: 1.4100 million CNY/year; 20 staffs with a salary 50,000CNY/year each, 41% of that is other welfare. =1.4100 million CNY/year (iii) Material: 15CNY/kW, i.e. 49.8MW*15CNY/kW=7.47 million CNY/10 year=0.747 million CNY/year (iv) Other fees: 30CNY/kW, i.e. 49.8MW*30CNY/kW=1.4940 million CNY (see the table below) (*Other cost is a fund preparing for unexpected cost for O&M. It is generally reserved according to the scale of project. Other cost of the proposed project was reserved 30CNY/kW in IRR sheet. It is a common practice in the design of wind power project). (v) Insurance: 0.25% of the total static investment, i.e. 1.1697 million CNY Total: 11.839 million CNY In addition, the annual O&M costs were reviewed during the on-site interview with the project owner. It was reported that the annual O&M costs would be even higher than the costs predicted in the FSR due to the increasing raw material prices and labour costs. Further more, the validation compared the Annual O&M Costs per Installation Capacity of similar projects in Shandong province which were registered in UNFCCC (refer to table 11), it is 0.12CNY/kWh for the proposed project, which is among the range from 0.08CNY/kWh to 0.21CNY/kWh. Hence, The annual O&M cost is verified by validation team and confirmed to be valid.	Total static investment (TSI)	52,886 CNY	Fixed Asset	52,886-6,097=46,789 6,097 is the equipment input tax	Maintenance fee	1.5%*Fixed Asset =46,789*1.5%	Annual value	7.0183 million CNY
Total static investment (TSI)	52,886 CNY										
Fixed Asset	52,886-6,097=46,789 6,097 is the equipment input tax										
Maintenance fee	1.5%*Fixed Asset =46,789*1.5%										
Annual value	7.0183 million CNY										
Project Lifetime	20	FSR /15/, Benchmark	The selected 20 year of project lifetime is consistent to the value in Chapter 1 14.2~14.3 of FSR								

(Year)		Reference /41/	According to the benchmark reference /51/, project lifetime is clearly indicated as 20-years for financial analysis. By further reviewing the "Guidance on the Assessment of Investment Analysis (Version 05)" EB62, it is written that in general a minimum period of 10 years and a maximum of 20 years of financial project lifetime will be appropriate. Therefore, the selection is reasonable.
Bank Loan Rate	6.40%	FSR /15/, Website of the People's Bank of China /58/	The FSR was finished in December, 2010 and the long-term loan interest ratio was 6.40% issued by the People's Bank of China. The validation team confirms the ratio used in the FSR is valid.
Value added tax (VAT) (%)	17%	FSR /15/, government officials interview of OSV	The validation team is able to confirm the applied tax rate of VAT is 17% for wind power industry according to the State Tax Laws (CAISHUI[2008]156/57/) and it is applicable to the project.
Tax of City Construction & Maintenance (% of VAT)	5	FSR /15/, State Tax Law ³	The validation team is able to confirm the applied tax rate for City Construction and Maintenance is valid and consistent with the State Tax Law.
Education Tax (% of VAT)	4	FSR /15/, State Tax Law ⁴ , /59/	The validation team is able to confirm the applied education tax of 4% of VAT in Shandong Province is valid and consistent with the State Tax Law (No.448) of the State Council, the amendment of "Interim regulations of Educational Surtax" and "Interim regulations of Local Educational Surtax in Shandong Province".
Income Tax (%)	25	FSR /15/, State Tax Law ⁵	The validation team is able to confirm that current corporation income tax rate of 25% has been effective since 1 st January 2008.
Residual value (%)	5%	FSR /15/, Bench mark reference /51/	The residual rate is shown in FSR (i.e. Cashflow chart) and is 5% of fixed asset.

In addition, the validation team also reviewed the tariff, annual operation cost and total static investment of the project activity by crosschecking the corresponding input values from the other registered wind farm projects in the same region. The parameters applied in the other projects are tabulated as following:

Table 11: Comparison of the Project activity with similar registered CDM wind farm projects in Shandong Province (source: UNFCCC webpage)

3 State Administration of Taxation, National VAT law, (<http://www.chinatax.gov.cn/n480462/n480513/n480919/index.html>)

4 Order No.448 of the State Council, the amendment of "Interim regulations of Educational Surtax" and " Interim regulations of Local Educational Surtax in Shandong Province, <http://www.qdedu.gov.cn/qdedu/6dadb12c-bed5-462d-9c49-bd07e8082212/29dab101-0fe1-443d-89ca-591172f9aed7.html>

5 State Administration of Taxation, Corporation Income Tax Law (Ref.: President of the People's Republic of China Order [2007] no.63), (<http://202.108.90.146/guoshui/action/GetArticleView1.do?id=43127&flag=1>)

UNF CCC Ref. no.	Name	Installe d capacity MW	Total static Investm ent (million CNY)	Annual feed-in electricity (MWh/ye ar)	Bus-bar tariff (Inc. VAT) (CNY/kW h)	Annual O&M Costs/ Installed capacity (CNY/kWh)	Investment cost/installe d capacity (CNY/kW)	Annual Utilizatio n hours (h)
1008	Rongcheng Dongchudao Wind Farm	15.0	173.41	28,529	0.61	0.08	11,560.67	1,902
1010	Laizhou Diaolongzui Wind Farm	48.8	450.14	96,720	0.61	0.08	9,224.18	1,982
1019	Qixia Tangshanpeng Windfarm Project	21.8	162.00	36,580	0.61	0.21	7,431.19	1,678
1090	Shandong Changdao 27.2 MW Wind Power Project	27.2	268.68	58,154	0.61	0.11	9,877.94	2,138
1128	Shandong Weihai 69 MW Wind Power Project	69.0	705.11	140,553	0.61	0.13	10,218.99	2,037
1755	Roncheng Wind Power Project, 48.75MW	48.8	502.00	88,310	0.61	N/A	10,286.89	1,810
1789	Shandong Tuoji Island Windfarm Project	12.0	101.92	23,628	0.61	0.11	8,493.33	1,969
2019	Shandong Luneng Dongying 48MW Wind Power Project	48.0	520.58	95,136	0.61	0.16	10,845.42	1,982
2021	Shandong Haiyang Qiwershan Wind Power Project	15.0	154.03	28,890	0.61	0.13	10,268.67	1,926
2094	The 48MW wind power project in Lubei, Shandong Province	48.0	539.05	98,520	0.61	0.04	11,230.21	2,053
2397	Shandong Penglai Pingdingshan Wind Farm Project	48.0	582.70	101,904	0.61	0.11	12,139.58	2,123
2436	Guohua Dongying Hekou 49.5 MW Wind Farm Project (Phase 1)	49.5	507.58	102,396	0.61	0.14	10,254.14	2,069
2438	Guohua Binzhou Zhanhua 49.5MW Wind Farm Project (Phase 1)	49.5	479.77	99,324	0.61	0.15	9,692.32	2,007
2442	Guohua Dongying Lijin 49.5 MW Wind Farm Project (Phase 1)	49.5	510.32	103,140	0.61	0.14	10,309.49	2,084
2530	Shandong Laizhou phase I Wind Power Project	49.5	516.30	96,080	0.61	0.11	10,430.30	1,941
2584	Shandong	49.5	639.24	95,978	0.61	0.08	12,913.94	1,939

	Dongying 1st phase Wind Power Project							
2764	Yantai Dongyuan Laizhou 48.5 MW Wind Farm Project Phase I	48.5	449.24	88,187	0.61	0.16	9,262.68	1,818
2814	Shandong Rushan Luneng Wind Farm	42.0	389.39	69,125	0.61	0.13	9,271.19	1,646
2831	Huadian Laizhou Wind Farm Project	40.5	425.56	77,320	0.61	0.06	10,507.65	1,909
3353	Huaneng Changyi Phase I Wind Farm Project	49.5	624.48	111,875	0.61	0.08	12,615.76	2,260
3391	Shandong Huaneng Shouguang 49.5MW Wind Farm Project	49.5	559.07	96,479	0.61	0.13	11,294.34	1,949
3603	Huaneng Binhai Wind Farm Project	49.5	630.35	114,114	0.61	0.08	12,734.34	2,305
3616	Yantai Dongyuan 25.5 MW Wind Farm Project Phase I	25.5	251.95	51,689	0.61	0.20	9,880.39	2,027
3718	Shandong Huaneng Hekou Phase I Wind Farm Project	49.5	545.31	100,090	0.61	0.12	11,016.36	2,022
4402	Shandong Penglai Wind Farm Phase II Project	48.0	539.62	103,740	0.61	0.10	11,242.08	2,161
4587	Penglai Daliuhang Wind Farm Project Phase I	49.8	550.55	103,883	0.61	0.08	11,055.22	2,086
4612	Weihai Jingqu 49.8 MW Wind Farm Project Phase I	49.8	541.95	102,986	0.61	0.08	10,882.53	2,068
4614	Huaneng Laizhou Phase II Wind Farm Project	49.5	509.29	89,100	0.61	0.14	10,288.69	1,800
4699	Huaneng Shandong Rongcheng Chudao Wind Power Project	48.0	492.61	90,240	0.61	0.15	10,262.71	1,880
4830	Huaneng Shandong Hekou Phase III Wind Farm Project	49.5	549.00	100,023	0.61	0.13	11,090.91	2,021
4871	Huaneng Shandong Hekou Phase IV Wind Farm Project	49.5	545.81	99,488	0.61	0.13	11,026.46	2,010
4882	Guohua Rongcheng Phase II Wind Farm Project	49.5	535.95	102,337	0.61	0.04	10,827.27	2,067
4923	Shandong Binzhou Zhanhua Tao'erhe Wind Power Project	49.5	512.13	96,525	0.61	0.15	10,346.06	1,950
5006	Shandong	38.25	390.51	71,750	0.61	0.14	10,209.41	1,876

	Liuwangzhuang Wind Farm Project							
5081	Huaneng Shandong Hekou Phase II Wind Farm Project	49.5	557.51	101,351	0.61	0.13	11,262.83	2,047
5252	Huaneng Shandong Shouguang Phase II Wind Power Project	49.5	562.757	96,030	0.61	0.12	11,368.83	1,940
5269	Huaneng Shandong Shouguang Phase III Wind Power Project	49.5	563.39	96,676	0.61	0.12	11,381.62	1,953
	Average /or range					0.13	10,621.75 Max: 12,913.94 Min: 7,431.19	Max: 2305 Min: 1646
	Proposed project	49.8	528.86	99,026	0.61	0.12	10,619.68	1988

The validation team also reviewed the parameters applied in other registered wind power projects in the same region (37 nos.) for the comparison with the proposed project activity. It is found that the total static investment per installed capacity of the project activity (10,619.68 CNY/KW) is within the range of the other registered projects (7,431.19~12,913.94 CNY/KW). For the ratio of annual O&M cost per kWh, the result is between the range of 0.04 CNY/KWh to 0.21CNY/KWh. the ratio of O&M per KWh for the project is 0.12CNY/KWh, it is lower than the average of 0.13CNY/KWh, Thus the validation team considers that the estimation of total static investment, O&M cost, Grid-In tariff are reasonable and conservative in the investment analysis.

Fixed values of financial parameters being applied

Fixed values of financial parameters that were applied throughout the whole financial period were validated by the validation team. By reference to the applied benchmark source /41/, it stated that “financial analysis is based on current national financial regulations and price system. For electric engineering projects, price level in the construction year can be applied to calculate project expense and revenue” (i.e. Chapter 3.1.2 of the benchmark /41/). The validation team validates that all the costs have been kept constant. Therefore, the validation team confirms that the fixed values of the financial parameters that were applied are regulated by the applied benchmark for financial analysis.

Highest tariff consideration for conservative approach (for reference only)

The validation team observed that the highest tariff utilized by the registered projects in UNFCCC website till Oct. 2011 in Shandong Province is 0.61CNY/kWh (incl. VAT) /62/. It is also observed that the estimated tariff of the proposed project (i.e., CNY0.61) is the highest tariff. The project IRR is 5.63%, below the benchmark IRR (i.e. 8% of project IRR).

From the above, it can be concluded that the financial analysis is confirmed to be accurate and traceable. In addition, the calculation in the financial spreadsheet based on the above parameter was checked, and confirmed to be accurate.

As indicated in the PDD/10/ and IRR sheet /11/, Validation team confirms that without the revenue from CDM, the project IRR would be 5.63%, which is below the benchmark IRR of 8%. Hence, the project cannot be considered as financially attractive in the absence of CDM benefits.

Sensitivity Analysis:

Validation Opinion: A sensitivity analysis is presented in the PDD which demonstrates that the project activity is unlikely to be financially viable under reasonable variations in the critical assumptions, i.e. fluctuation range of $\pm 10\%$, in four selected financial parameters, incl. (i) Total static investment, (ii) Annual O&M cost, (iii) Annual net electricity supplied, and (iv) Electricity tariff. The selection of the sensitivity test parameters focuses mainly on those parameters which are considered more likely to vary during the project implementation with reference to the "Guidance on the Assessment of Investment Analysis (Version 05)" EB62. It states, "The parameters analysed shall at least include that which contribute more than 20% of either total project costs or total project revenue".

The selection of the fluctuation range of $\pm 10\%$ is also consistent with the statement of the guideline. The validation team observed that the Project IRRs would not cross the benchmark when the total static investment and, tariff (Incl. VAT), annual net electricity output fluctuated to $+10\%$, and the Project IRR would not cross the benchmark when the O&M cost decrease to -10% , the project IRR is still lower than the benchmark, i.e. project IRR of 8%.

In the PDD, the circumstances of change of level in each parameter, that will enable the project IRR to reach the 8% has been presented and analyzed. The validation team has checked each parameters as follows;

- *If the total static investment is decreased by 16.2%, the project IRR can reach to the industrial benchmark.*

This is very unlikely to happen: By comparing the expenditure from the Contract List/21/ (includes, i.e.: the signed equipment contracts /25/ /26/ and Construction contract /33/, etc.) show that the actual costs summed in the Contract List/21/ account for 90.11% of the estimations of FSR. Validation team can confirm that the actual total investment is unlikely to decrease. In addition, the raw material price for wind power facilities was booming in recent years in China /66/. It is impossible to decrease the budget in FSR. Therefore, it seems unlikely for the static investment to decrease by 16.2% and the project IRR will not reach 8%.

- *If the annual O&M cost decreases 73.8%, the project IRR may reach the industrial benchmark.*

According to the statistics /61/ published by the National Bureau of Statistics of China, the prices of the materials and industrial products, and labor cost have kept rising up in

the past several years. Therefore, it is unlikely for annual O&M cost to decrease and the project IRR will not reach the industrial benchmark of 8%.

- *If the annual net power supply increased by 17.7%, the project IRR can reach to the industrial benchmark.*

It is confirmed that the power generation is calculated based on the survey data of the local wind resources from year 1979 to 2008. The annual generation represents a long-term average power supply throughout the lifetime of wind farm, where the yearly-variations have already been taken into account. Hence, the significant increase in annual power supply is unlikely to happen;

- *If the electricity tariff rate increased by 17.7%, the project IRR can reach to the industrial benchmark.*

By reviewing the estimated tariff during the investment decision making, it applied <Notice on Improvement the tariff of Wind Power Project (FaGaiJiaGe [2009] No.1906)> in which the tariff was determined 0.61CNY/kW(VAT incl.) in Shandong Province; In addition, the on-grid tariff for wind power project is strictly regulated by the Chinese government, and the tariff adjustment is a national policy, referring to the document hence the tariff is unlikely to experience dramatic changes in the near future. Therefore, it is deemed that it is unlikely that the tariff will increase by 17.7%. And the project IRR will not reach to 8%.

Therefore, based on provided information and the validation of various references, the project is proved to be "not the most financially attractive" and the project is additional.

3.5.4 Barrier analysis

- Not applicable as barrier analysis is not applied.

3.5.5 Common practice analysis

Validation Opinion: According to PDD, there is only one similar wind farm project was identified for common practice analysis under the following criteria: (i) Shandong Province with (ii) installed capacity more than 15 MW (iii) starting to construct after 2002; and (iv) not applying or registered as CDM projects are identified for the common practice analysis in the PDD.

The validation team agrees that the chosen geographical scope (i.e. Shandong Province) was appropriately and conservatively determined, considering that all the identified projects that take place in Shandong Province share a comparable environment with respect to regulatory framework, investment climate, access to technology and financing, and other local policies, including tariff assessment/approval procedure and policies, etc.

The selected range of the installed capacity (more than 15MW) is also considered appropriate as it covers the analysis of wind power projects within a similar project

scale. According to the description for the comparable size to the project activity approved by EB (i.e., less than 15MW is a small scale project), hence, a similar scale to the proposed project is defined as more than 15MW.

Moreover, it is justifiable to include the projects commissioned since the Year 2002 in the analysis, because there was a national power policy change in 2002 in the P.R. China, where the power plants and power grids were built independently. This implied a distinctive difference for projects built before 2002 from the proposed project; as they were all developed by stated-owned enterprises, or constructed with national or local government funds, or the government provided the loan guarantee, etc., for ensuring their project implementation.

According to the *Statistics of Wind Turbine Installation in China 2010* /64/ totally 1418.7MW wind turbines are installed in Shandong Province up to the end of 2010 /50/. There are totally 38 similar wind farm projects were detected, among them, 37 projects have been developed or are being developed as CDM projects to improve their financial status and decrease risks. The project named “Shandong Qingdao Huawei Wind Power Project” is an early demonstration project and received ODA support from DEG (part of KfW Bankengruppe)/63/. Hence, it can be distinguished from the proposed project. The validation team has checked the reference source, and confirmed that the reference source, which was publicly available, provided the best available information to the PP for conducting the common practice analysis. All listed wind power projects in the PDD were consistent with the projects in the reference source. So the proposed project without support of CDM is not common practice in Shandong Province in China.

Hence, the analysis in the PDD is considered to have comprehensively covered the similar project for discussion. Hence, the project is not a common practice in Shandong Province.

In summary, based on the provided information and description in PDD, the validation team confirms the project is additional.

3.6 Monitoring

Validation Opinion: The project proponent applies the approved consolidated monitoring methodology ACM0002/Version 12.2.0 – “Consolidated baseline methodology for grid-connected electricity generation from renewable sources” for the project activity. Applicability criteria of the monitoring methodology to the project activity are met as noted in Section 3.4 of this report.

The project activity will neither generate any project emission nor leakage in accordance with ACM0002/Version 12.2.0 (Detailed discussion on calculation of project emission and leakage have been addressed in Section 3.4 above). Therefore monitoring of project emission and leakage is not required.

The monitoring will involve measuring the electricity generated by the project and electricity supplied from the power grid (i.e. NCPG) to the project. The CM emission factor is determined *ex-ante* based on the most recent information available during the PDD submission for DOE for validation. According to the methodology and calculation tool, CM emission factor during the first crediting period remains unchanged. Monitoring of GHG emission reduction is based on determining the net quantity of electricity supplied to the NCPG. The PDD has made provisions in B.7 for full details in monitoring the GHG emissions reduction due to the project activity.

3.6.1 Parameters determined ex-ante

Validation Opinion: The parameters for determining the GHG emission reductions have been clearly demonstrated in B.6.2. of PDD. The validation team has verified those parameters by means of document review (i.e. FSR /15/, tables of IPCC 2006 /43/, China Electric Power Yearbooks /53/, China energy Statistics Yearbooks /54/, NDRC Emission factors /52/ and methodological tool /5/), and interviews during OSV. Please refer to section 3.4.4 of this report for details. Therefore, validation team can conclude that all relevant parameters to calculate the GHG emission reductions of the project have been sufficiently considered and the value of the parameters is real, measureable.

3.6.2 Parameters monitored ex-post

Validation Opinion: According to the methodology ACM0002/ Version 12.2.0 and methodological tool, there are only one parameter that have to be monitored during the crediting period, it is “the quantity of net electricity generation supplied by the project activity to the grid in year y”. This parameter is described in details at section B.7 of PDD.

According to the approval of Grid-In Agreement /45/ and the OSV /vi/ /vii/, the main meter (M1) will be installed in the Tangtian Substation, and the back-up meter(M2) will be installed at the outlet of the 35kV/110kV substation, M3 is also located at the outlet of the 35kV/110kV substation is measuring the electricity generated by Project phase II, and M4 is located in the plant in the spare 10kV line from the Grid. And the accuracy of the meters M1, M2, M3 is not less than 0.5s. The accuracy of M4 is not less than 2.0.

The net power supply by the proposed project to grid is monitored, computerized from the monitoring parameters ($EG_{facility,y}$), the measurement of generated electricity is continuously and recorded monthly which meets the methodology requirement.

The proposed project will share the transformer and gateway meters with Qingdao Longxin Wind Power Project Phase II, the following formula is applied to get the net electricity supplied to the grid by the proposed project:

$$EG_{facility,y} = EG_{output,y} - EG_{another,y} - EG_{import,y} - EG_{im-spare,y}$$

Where:

$EG_{facility,y}$: Net electricity supplied to the Grid by the proposed project in the year y.

$EG_{output,y}$: Total electricity supplied to the Grid by the proposed project and Phase II project in the year y, which will be measured by bidirectional meter M1 installed in the Tangtian substation;

$EG_{import,y}$: Total electricity imported from the Grid by the proposed project and Phase II project in the year y , which will be measured by bidirectional meter M1 installed in Tangtian substation;

$EG_{im-spare,y}$: Electricity imported from the Grid through a spare 10 kV line in the year y , which will be measured by meter M4 installed on the spare power line on the site of the proposed project

$EG_{project,y}$: Electricity generated by the proposed project in the year y , which will be measured by electricity meter M2;

$EG_{another,y}$: Electricity supplied to the grid by the Phase II Project in the year y , which will be measured by electricity meter M3;

$EG_{import,y}$ is the total electricity imported from the Grid by the proposed project and the Phase II project in year y , to be conservative, when the $EG_{facility,y}$ is calculated, $EG_{import,y}$ is fully deducted including the electricity imported from the Grid by Phase II project. Validation team confirms it is valid.

Table 12: the meters outlined in the monitoring plan

Meter	Location	Function	Owner	Precision	Calibration Frequency
M1	Tangtian Substation	Electricity supplied to and backup from the Grid Co by phase I and II.	CRWQC	Not lower than 0.5s	Once/Year
M2	On site substation	Measure the electricity supplied by Phase I	CRWQC	Ditto	Ditto
M3	On site substation	Measure the electricity supplied by Phase II	CRWQC	Ditto	Ditto
M4	Project site	Measure the electricity used by the project through spare 10kV line.	CRWQC	No less than 2.0	Once per 3 years

The validation team confirms that the above formula is practical, and the manner is valid. And the validation team considers that the monitoring plan has complied with requirement in the approved methodology.

3.6.3 Management system and quality assurance

Validation Opinion: Monitoring of sustainable development indicators is not required by the China's DNA. The environmental impacts are considered minor and will be monitored by the local environmental authority during the project lifetime. According to the PDD, the project's monitoring plan outlines the followings:-

- Monitoring object which comprises the monitoring plan will provide credible, accurate, transparent and conservative monitoring data;
- Monitoring Structure, which comprises the mains roles and responsibilities of the CDM Team;
- Monitoring Program and Equipment mainly defines the meter location, precision and calibration issues; it also indicated the formula to get the $EG_{facility,y}$.
- Monitoring Training Arrangement;
- Data collection and staff training;
- Calibration of Meters & Metering;
- Data Management System: with procedures for recording;
- Monitoring Report Preparation.

Referring to the PDD, the calibration frequency of the meters (M1, M2 and M3) is once per year and M4 is once per 3 years during operational period. Validation team confirms it is valid.

The roles and responsibilities for monitoring have been identified. The procedure for QA/QC is also discussed in details to ensure the electricity data measured and recorded via cross-checking the sales receipt and back-up meter.

The validation team validated the monitoring plan of the project by document review /10/ /15/ and interviews with PP /vi/ /vii/ and CDM consultant /ix/~x/ confirmed that the monitoring plan written on PDD could reflect the actual measurement and on-grid arrangement. Based on the observation and interviews with the PP during OSV, the validation team considered that the PP has ability to implement the monitoring plan under the assistance of CDM consultant, i.e. Easy Carbon Consultancy Co., Ltd.

3.7 Sustainable Development

Validation Opinion: The validation team validated that the project is considered to be contributing to sustainable development (SD) in the host country, by on one hand utilizing renewable wind power resources available in the project region, and on the other hand eliminating the environmental pollution caused by operation of fossil fuel-fired plant. In addition to CO₂ emission reductions, the project would mitigate other pollutants, such as SO₂, NO_x and particulates associated with power generation from fossil fuel. SD is further confirmed from the LoA issued by the DNA of P.R. China, i.e. NDRC /68/. Other benefits would be expected by implementation of the proposed project, which includes improvement in the local and regional economic development by generating additional tax income from electricity sales of the proposed project to local government. Furthermore, the project could provide a few temporary and permanent employment opportunities to the communities especially for the workers from local area. The above mentioned social benefits were confirmed by validation team during OSV. As confirmed by the local government officials (/i/ /ii/) during on-site interview, sustainable development in social, environmental and economic aspects will be achieved by implementation of the proposed project.

3.8 Environmental Impacts

Validation Opinion: The environmental impacts (including air, noise, solid waste, ecosystem) have been reported in the Section D of PDD. It is reported that the project activities are not expected to cause significant environmental impacts in general. The environmental impacts of the project were sufficiently assessed by means of an Environmental Impact Assessment Report (EIAR) study according to the China laws & regulations. The Environmental Impact Assessment Report (EIAR) /29/ which is prepared Qingdao City Environmental Protection Science Research Institute in September 2010, has been approved by Environmental Protection Bureau of Qingdao City on 12th Oct. 2010 /30/. These documents were checked by the validation team and confirmed to be valid.

During OSV, no significant environmental impact was identified from the project activities since relevant mitigation measures will be properly implemented as stated in the EIAR/29/. The environmental mitigation and prevention measures are satisfied as confirmed by the interview with representative of local EPB /ii/ and local villagers /iii~/v/. The representative of local EPB /ii/ was interviewed and confirmed that since project sites are far from the residents (at least 5 km), the influence from the construction are very rare to the residents. The proposed project doesn't cause any villager resettlement; and the project did not occupy any cultivating land as confirmed from local officials/i/ /ii/ and villagers /iii~/v/. Hence, there is no significant impact on the land acquisition. And there is limited impact on the ecosystem by the project according to both the EIAR/29/ and the official from the local environmental protection bureau/ii/.

3.9 Local Stakeholder Consultation

Validation Opinion: It is not a formal requirement by the current legislation of the host country that a construction project which only requires EIAR, has to carry out a public consultation process. One public consultation was conducted for the project in March 2011 by the project owner.

Total 50 sets of questionnaires were distributed and 49 of them were received. Questionnaires were checked by validation team during OSV and confirmed valid.

The background of the stakeholders such as age, sex, education and nationality were reported and their comments were also summarized in the Section E of PDD. Most of the comments from stakeholders considered that the project would bring positive impacts to the local economy and livelihoods of local people with increased job opportunities and tax income to local community. The majority of them expressed their supports to the implementation of the project activity. There is no adverse opinion on the project from the public consultation.

The validation team also checked the stakeholder comments during OSV. The representatives /iii~/v/ from the local community were interviewed. In general, the interviewees showed adequate understanding of the nature of the project and were satisfactory with the arrangements on the public concerns, such as the dust, noise control, etc. As confirmed by the government officials from local EPB /ii/ and local DRC /i/ during OSV, the residents nearby the project location consider that the project would benefit the improvement in local social, economic and environmental development. The response is overall supportive to the project implementation. Therefore, the validation team can conclude that that public consultation is effective to reflect the voice of affected stakeholders.

3.10 Comments by Parties, Stakeholders and NGOs

Validation Opinion: The PDD/Version 1 of 12th July 2010 was made publicly available on UNFCCC's website:

(<http://cdm.unfccc.int/Projects/Validation/DB/WEYY5DMTUSY5J5GLVWH0NVQ7AW88KL/view.html>)

And parties, stakeholders and NGOs were through the CDM website invited to provide comments during a 30 days period from 10th September 2011 to 9th Oct. 2011, where no comment was received.

Table 1: Validation requirements

(Based on § 37 of the CDM Modalities and Procedures and on CDM Validation and Verification Manual, Annex 1 of EB55)

Checklist question	Ref.	MoV ⁶	Findings, comments, references, data sources	Draft conclusion	Final conclusion
1. Approval					
1.1 Have Letters of Approval have been provided from all involved Parties? If yes, indicate: – when and by which Party the LoA has been issued, with a clear reference to the LoA itself and any supporting documentation; – whether the LoA was provided to the DOE by the project participants or directly by the DNA; – the means of validation employed to assess the authenticity of the document; and – by a clear statement, that the DOE considers the LoA to be valid.	LoA	DR, WWW	This is a unilateral project, LoA from the Host DNA is not provided, corrective action is needed. CAR1 Please submit the LoA of the host country for validation when available.	CAR1	OK
1.2 Are all Parties, who issued the LoA, Parties to the Kyoto Protocol <u>and</u> is this stated in	LoA	DR, WWW	China is a party to Kyoto protocol since 16/02/2005 (UNFCCC website).	See 1.1	OK

⁶ MoV = Means of Verification, DR = Document Review, I = Interview, www = internet search.

the LoA?					
1.3 Is every LoA from the Parties involved issued by an organisation listed as Designated National Authority (DNA) on the UNFCCC web site? <i>Indicate the official name of the DNA and contact person name.</i>	LoA	DR, WWW	The LoA of the host county is missing.	See 1.1	OK
1.4 Is the participation in the CDM project activity voluntary <u>and</u> is this stated in all LoAs? <i>Indicate the source of proof.</i>	LoA	DR	The LoA of the host county is missing.	See 1.3	OK
1.5 Is the LoA unconditional with respect to 1.2 to 1.4?	LoA	DR	The LoA of the host county is missing.	See 1.3	OK
1.6 Is the title of the CDM project activity as given in the PDD identical with the title given in all LoAs and Modalities of Communication? <i>Provide Yes/No answer, and include details into Tables 2, 3 and 4 accordingly.</i>	LoA, MoC	DR	As a unilateral project, no MoC is provided.	See 1.1	OK
1.7 If any of provided LoAs contains additional specification of the CDM project activity (PDD version number, validation report version number, amount of ER,	LoA, /10/	DR	The LoA of the host county is missing.	See 1.1	O.K.

etc.) are those specifications valid and consistent with other documents?					
1.8 Does the project activity involve any public funding from Annex I Parties? <u>If yes</u> , has Annex I Party provided a written confirmation that the use of such funding does not lead to the diversion of the official development assistance.	/10/, /16/	DR, I	No. After communicating with the representative of the project owner, Mr. Wei Donglin and reviewing the approval of FSR, the validation team confirms around 30% of the total investment will be raised from equity, and the rest 70% will be financed from bank loan. Consequently no any public funding from annex1 party is involved.	O.K.	O.K.
2. Participation (VVM E.2)					
2.1 Are the Parties and project participants (PP) listed in the section A.3 of the PDD correctly <u>and</u> is this information consistent with the contact details provided in Annex 1 of the PDD?	/10/, LoA	DR, I	The LoA of the host county is missing.	See 1.1.	O.K.
2.2 Has every Party involved approved the participation of each corresponding PP, either by means of a LoA or by a separate written document? <i>Indicate Yes / No answer and describe all inconsistencies in the Tables 2, 3 and 4 accordingly.</i>	LoA	DR, I	The LoA of the host county is missing.	See 1.1 and 1.3	O.K.
3. Project Design Document (VVM E.3)					

3.1 Is the PDD presented for validation based on the latest template available at the UNFCCC website? <i>Indicate Yes / No answer and describe all inconsistencies in the Tables 2, 3 and 4 accordingly.</i>	/10/ , UNFCCC	DR	Yes. Version 03 of PDD template is the latest version available on the UNFCCC's website: http://cdm.unfccc.int/Reference/PDDs_Forms/PDDs/PDD_form04_v03_2.pdf	O.K.	O.K.
3.2 Has the PDD been established in accordance with the CDM requirements for completing PDDs issued by the CDM EB?	/10/ UNFCCC	DR	Yes. Having reviewed the PDD, the validation team confirms PDD is established as per the Guidelines for completing PDD.	O.K.	O.K.
4. Project Description (VVM E.4)					
4.1 Does the PDD contain a description, which provides the reader with a clear understanding of the precise nature of the project activity and the technical aspects of its implementation? 4.1b) Is the description (incl. any process flow-charts, Spreadsheets etc.) complete, coherent and consistent with the provisions of the monitoring plan?	/10/, /15/,	DR,I	No, PDD contains a description but not clear, complete and accurate enough, or several errors is found, thus CARs are needed. CAR2 Please describe the detailed project location in A.4.1.4. From the map, we can not see the Longxin as a location. Please demonstrate the technical training arrangement of the Operation and Maintenance of the plant. Please submit the layout of the project and the project schedule. Please provide the Project IRR sheet and ER sheet to Validation team. CAR3 PP is requested to indicate the type of turbines in A.4.3 of PDD, and fill the key events on the	GL01 CAR2 CAR3 CAR4	O.K.

		timeline in table in the B.5 in the PDD. Please show to validation team the complete version of the FSR (incl. Cover and second page) which was to submit to DRC for approval. Please submit the land use approval for the project. CAR4 The format of the geographical coordinates should meet the following requirement: – All GPS coordinates indicated in the PDD should be presented in the decimal format with +/- sign only. Coordinates with multiple types of specifiers will not be accepted. – All GPS coordinates should have a precision of 4 (four) decimals. Lower precision will not be accepted. Higher precision will have to be rounded up, otherwise will not be accepted. - GPS coordinates for the project implementation area (ex. square, triangle, polygon,etc.) in the PDD; - Single set of GPS coordinates in the PDD, which would depict an approximate geometrical middle/center point of the area of the implementation; - A table with a set of coordinates for each implementation point. CL1 Any loan contract signed, if yes, please provide scanned copy to validation team. Please show evidence that the project is not a		
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			project of ODA involved in. Please provide the current organization chart of the Company. Please provide the CDM consulting contract of the project to validation team.		
4.2 In the case of greenfield project activity, is the project design described sufficiently by means of specifications, drawings and manuals? <i>Provide Yes/No answer and indicate the documents which have been reviewed in relation to the issue.</i>	/10/, /15/, /16/	DR	Yes, According to FSR and its approval, the project is a greenfield (newly-built) wind power station, which falls into the scope 1, renewable energy. The validation team can confirm that the project design is described by means of the above mentioned documents and it is sufficient.	O.K.	O.K.
4.3 Does the project activity reflects current good practices, uses state of the art technology or would the technology result in a significantly better performance, than any commonly used technologies in the host country? <i>Provide the description of how validation has been carried out and what comparisons have been made.</i>	/10/, /15/, /16/	DR, I	Yes. The validation team reviewed the electricity power yearbook (2005 to 2009), in which it is showed that at present the electricity supply in China is dominated by the fossil fuel-fired thermal plants. While the proposed project utilizes renewable energy, i.e. wind power. It can mitigate the anthropogenic emission. Hence the project activity reflects current good practices. On the other hand the FSR of the proposed project has been completed by the qualified research institute and has been approved by the authorized government. The technology used in the project is considered as the state of the art in China.	O.K.	O.K.
4.4 In cases where the project activity involves the alteration	/10/, /15/, /16/	DR, I	N/A. As it is a newly-built hydro power project, no existing installation or process.	O.K.	O.K.

of an existing installation or process, does the PDD provide a clear description of the differences between the project and the pre-project scenario? <i>Please, provide Yes/Now answer and update Tables 2, 3 and 4 accordingly, if there is anything unclear in the provided description.</i>					
5. Baseline and Monitoring methodology					
5.1 General requirements					
5.1.1 Is the methodology used in the project activity approved by the CDM EB <u>and</u> is the selected version still valid?	UNFCCC, /10/	DR, WWW	Yes. The project applies ACM0002 (Version 12.2.0): Consolidated baseline methodology for grid-connected electricity generation from renewable sources , which is approved by CDM-EB and Valid from 17 Sep 2010 onwards.	O.K.	O.K.
5.2 Applicability of the selected methodology					
5.2.1 Does the project activity qualify under the criteria for small-scale CDM project activities set out in § 6 (c) of decision 17/CP.7 and Annex II of the Modalities and Procedures for the CDM? <i>Please provide Yes/No response and description of</i>	/10/, /15/, /16/	DR I	No. According to the FSR and its approval, and onsite check, the project activity is a renewable energy greenfield project whose total installed capacity is 49.8MW, which is higher than the 15 MW installation capacity set out in § 6 (c) of decision 17/CP.7 and Annex II of the Modalities and Procedures for the CDM. Thus, it is qualified under the criteria for large-	O.K.	O.K.

<i>how this was validated.</i> <i>In case of calculated emission reductions varying over time, SSC-applicability limits must be met for every single year in any of the max. 3 subsequent crediting periods.</i>			scale CDM project activities.		
5.2.1.1 If yes, does the PDD extensively demonstrates and confirms that the small-scale project activity is not a debundled component of a larger project? <i>Please indicate Yes/No answer. In case of positive conclusion provide details of the validation measures taken and data found during the procedure. Otherwise amend the Tables 2, 3 and 4 accordingly.</i>	/10/, UNFCCC,	DR	N/A. please see 5.2.1.	O.K.	O.K.
5.2.2 Are all applicability conditions of the selected baseline and monitoring methodology and all tools involved satisfied by the project activity? <i>Please indicate Yes/No answer. In case of positive conclusion provide details of the validation measures. Otherwise amend the Tables 2, 3 and 4 accordingly.</i>	/15/, /16/, /5/	DR	Yes. All the applicability conditions of the selected methodology and tools are satisfied by the project activity which are as follows: 1. The proposed project is a Greenfield wind power project and connected to the North China Power Grid (NCPG); 2. The geographic and system boundaries for the relevant electricity grid can be clearly identified (NCPG); 3. The proposed project does not involve fuel switching from fossil fuels.	O.K.	O.K.

5.2.3 Is the selection of the applied baseline and monitoring methodology justified?	/10/, Tools, /4/	DR	Yes, the selection was sufficiently justified in Section B.2 of PDD	O.K.	O.K.
5.2.4 Is the selected methodology correctly quoted in all related documents?	/10/, Tools, /4/	DR	Yes, the selected methodology, ACM0002 is quoted correctly in the PDD.	O.K.	O.K.
5.2.5 Does the PDD sufficiently describe all the GHG emission sources or sinks occurring as a result of project activity, which have not been accounted for under the selected methodology and are expected to contribute more than 1% of the overall expected average annual emission reductions? <i>Provide Yes/No answer. Indicate the sources or sinks of GHG, which were proved to be negligible. Otherwise amend the Tables 2, 3 and 4 accordingly.</i>	/10/, /15/	DR, I	No, CL is thus needed. CL2 Please clarify whether there is any possible GHG emission source or sink occurring as a result of project activity, which have not been accounted for under the selected methodology and are expected to contribute more than 1% of the overall expected average annual emission reductions. If yes, please consider it as per para 77 of VVM.	CL02	O.K.
5.3 Project boundary					
5.3.1 Does the PDD correctly describe the project boundary? <i>Provide Yes/No answer. And amend the Tables 2, 3 and 4, if needed.</i>	/10/, /12/	DR, WWW	Yes. The project spatial boundaries have been defined and are described in A.4 and B.3 of PDD. The project boundary is clearly defined as the project activity and power plants physically connected to the NCPG, which is composed by the sub-grid of Beijing, Tianjin, Shanxi Province, Hebei Province, Shangdong Province, Inner Mongolia Autonomous Region.	O.K.	O.K.

			So the project boundary is the geographic extent of NCPG.		
5.3.2 Does the PDD correctly indicate and describe the emission sources and sinks of GHG gases that are included in the project boundary?	/10/ /4/	DR	Yes, it is clearly described in the table in chapter B3 of the PDD. According to the approved methodology ACM0002 version 12.2.0 the CO2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity is the main emission source.	O.K.	O.K.
5.3.3 In cases where the methodology allows project participants to choose whether a source or gas is to be included in the project boundary, is the choice explained and justified by PPs?	-	-	Inapplicable.	N/A	N/A
5.4 Baseline identification					
5.4.1 Has the procedure contained in the selected methodology to identify the most reasonable baseline scenario been applied correctly and documented in the PDD?	/10/, /4/	DR	Yes. According to the methodology ACM0002, version 12.2.0, as the project activity is a greenfield project, the baseline scenario is indicated clearly: If the project activity is the installation of a new grid-connected renewable power plant/unit, the baseline scenario is the electricity delivered to the grid by the project activity that otherwise would have been generated by the operation of grid-connected power plants and by the addition of new generation sources. Hence, the procedure contained in the selected methodology to identify the most reasonable baseline scenario has been	O.K.	O.K.

			applied correctly in the PDD.		
5.4.1.1 Is the identified baseline scenario plausible?	/10/, /4/	DR	Yes, the identified baseline scenario is plausible as indicated in the PDD.	O.K.	O.K.
5.4.1.2 Are all assumptions stated in a transparent and conservative manner?	/10/, /4/	DR	Yes, As the baseline scenario has been indicated clearly in PDD indicates that related assumptions are shown transparently and conservatively	O.K.	O.K.
5.4.2 Does the selected methodology require the use of tools and does PDD reflects that correctly?	/10/, /4/	DR	Yes, the ACM0002 was applied with related tools.	O.K.	O.K.
5.4.2.1 Were all the tools applied correctly?	/10/, /4/	DR	Yes, all the tools are applied correctly.	O.K.	O.K.
5.4.3 In case the methodology requires several alternative scenarios to be considered in the identification of the most reasonable baseline scenario, have all scenarios been considered and have no reasonable alternative scenario been excluded?	/10/, /4/	DR	Yes, it was corretly done accordingly.	O.K.	O.K.
5.4.3.1 Has the choice of the baseline scenario been done using conservative assumptions?	/10/, /4/	DR	Yes, it was done using condervative assumptions.	O.K.	O.K.
5.4.4 Is the identified baseline scenario reasonable according to the assumptions, calculations and rationales	/10/, /4/	DR	Yes, the identified baseline scenario is reasonable according to the assumptions, calculations and rationales used in the PDD.	O.K.	O.K.

used in the PDD and other reference sources?					
5.4.5 Does the PDD describe how the national and sectoral policies relevant to the baseline scenario have been identified and considered in the PDD?	/10/ /4/	DR	Yes, the national and sectoral policies relevant to the baseline scenario have been correctly identified and considered in the PDD	O.K.	O.K.
5.4.6 Does the PDD provide a verifiable description of the identified baseline scenario, including a description of the technology that would be employed and/or the activities that would take place in the absence of the project activity?	/10/ /4/	DR	Yes, the PDD provides a verifiable description of the identified baseline scenario, including a description of the technology that would be employed and/or the activities that would take place in the absence of the project activity.	O.K.	O.K.
5.5 Algorithm and/or formulae used to determine emission reductions					
5.5.1 Are all calculations applied and documented according to the selected methodology and in a complete and transparent manner? 5.5.1b) Are correct units applied and consistency between parameter dimensions and parameter value ensured? See also Question 4.1.b) with respect to consistency of parameter values between calculation spreadsheets and PDD.	/10/ /12/ /4/, /5/	DR	Yes. The validation team confirms all calculations are applied and documented as per the selected methodology in a complete and transparent manner except the emission factor and net electricity stated in the EF&ER worksheet. Moreover, the validation team checks the EF&ER spreadsheet and PDD, and confirms parameter values in spreadsheet complies with those in PDD. Also the correct units are applied in PDD and spreadsheet considering the corresponding parameter dimensions and value.	O.K.	O.K.
5.5.2 In case the methodology	/10/	DR	Yes, the validation team confirms adequate	O.K.	O.K.

allows a selection between different options for equations or parameters, has adequate justification been given and have the correct equations and parameters been used, in accordance with the methodology selected?	/4/ /5/		justification has been given and the correct equations and parameters have been used according to the ACM0002 version 12.2.0 except option 1 in step 2 of B.6.1 of PDD.		
5.5.3 In case some data and parameters will not be monitored throughout the crediting period, but have already been determined and fixed, are all data sources, assumptions and calculations correct, applicable to the proposed CDM project activity and conservative?	/10/ /4/	DR	Yes, the fixed parameter is the emission factor of NCPG. Having compared the emission factor applied in PDD with that published on the DNA of China's website on 20/12/2010, the validation team is able to confirm it is correctly applicable and conservative.	O.K.	O.K.
5.5.4 In case data and parameters will be monitored on implementation and hence become available only after validation of the project activity, are the estimates provided in the PDD for these data and parameters reasonable?	/10/,/15/	DR	Yes, according to the methodology ACM0002 version 12.2.0, the net electricity supplied to the grid is the monitoring parameters. Having reviewed the FSR and its approval separately, the validation team can confirm the estimate of the net electricity supplied to the grid provided in PDD is appropriated.	O.K.	O.K.
5.5.5 Have the major risks and uncertainties, which can influence the emission reduction estimates, been identified and addressed in the PDD?	/10/, /15/, /16/	DR	No, CAR is needed. CAR5 Please depict if there are some risks and uncertainties, which can influence the emission reduction estimates. If yes, please	CAR5	O.K.

			identify and address it in PDD.		
5.6 Leakage					
5.6.1 Has the leakage been identified and calculated according to the approved methodology?	/10/, /4/ /15/, /25/ /26/	DR	N/A according to the methodology ACM0002 version 12.2.0, no leakage are considered.	N/A.	N/A.
5.6.2 Have the leakage been addressed in complete, conservative and substantiated manner?	N/A.	DR	N/A. please refer to 5.6.1.	N/A.	N/A.
5.6.3 Are uncertainties in the leakage emission estimates properly addressed?	N/A.	DR	N/A. please refer to 5.6.1.	N/A.	N/A.
6. Methodology-related issues for afforestation or reforestation CDM project activities					
Add specific A/R requirements – if applicable!			Not applicable for this CDM project activity	O.K.	O.K.
7. Additionality					
7.1 Prior consideration of the CDM (VVM E.6.III.a)					
7.1.1 Is there documented evidence provided by the project participants on how and when the decision to proceed with the project activity was taken?	/10/, /17/	DR, I	Yes. The board meeting was held and company decision made on 15/03/2011 by the project owner, i.e. China Resources Wind Power (Qingdao) Co., Ltd. And the approval of the project issued on 02/03/2011 by the DRC of Qingdao City are provided.	O.K.	O.K.

7.1.2 Is the starting date of the project activity, reported in the PDD, in accordance with the “Glossary of CDM terms” and CDM VVM (§97)? Note: Confirm the starting date indicated in C.1. is consistent within the PDD, in particular with respect to the project implementation history.	/10/ /26/ /27/ /28/	DR, I	Yes. The starting date of the project activity is defined as the earliest date of actual initiation of the project activity. During onsite observation, the validation team checked the project site and communicated with the project owner, Mr. Wei Donglin and would like to confirm the construction of project activity started on 30 March 2011, which is the turbines purchasing time, and it is the earliest day of real action. CL3 Why the submitted contracts did not include the construction of powerhouse or admin house?	CL3	O.K.
7.1.3 Is the date stated in the provided evidence consistent with other available evidence (e.g. dates of construction, purchase orders for equipment)?	/10/ /26/ /27/ /28/	DR, I	Refer to 7.1.2.	Refer to 7.1.2.	O.K.
7.1.4 If the project was not published and the starting date is on or after 2nd August 2008, was it possible to receive from UNFCCC secretariat and/or DNA a written confirmation that PPs previously informed the above entities on commencement of the project activity and of their intention to seek CDM status?	/32/ /37/	DR, I	Yes. Since the project activity was published on 10/09/2011 while the project was started in March 2011, the pp notified the UNFCCC on 17th Oct. 2011/37/, hence, this requirement is achieved.	O.K.	O.K.

7.1.5 For the project activities with a starting date before 2 nd August 2008 and before the actual publication, was there enough evidence presented to prove that PPs were previously aware of CDM?	/10/, /26/ /27/ /28/	DR	N/A. For the reason that the start date is after 02/08/2008.	N/A.	N/A.
7.1.6 For the project activities with a starting date before 2 nd August 2008 and before the actual publication, was there enough evidence presented to prove that CDM benefits have been a decisive factor in the decision to proceed with the project activity?	/10/, /26/ /27/ /28/	DR	N/A. For the reason that the start date is after 02/08/2008.	O.K.	O.K.
7.1.7 Does the individual or body that took the decision to proceed with the project activity have/had the authority to do so?	/10/, /13/ /20/	DR	Yes. Having reviewed the business licence and the meeting records of prior CDM consideration, the validation team confirms this decision in the meeting records is valid legally.	O.K.	O.K.
7.1.8 For the project activities with a starting date before 2 nd August 2008 and before the actual publication, was there enough evidence presented to prove that PPs were taking continuing and real actions to secure CDM status for the project in parallel with its implementation?	/10/, /26/ /27/ /28/	DR	N/A. For the reason that the start date is after 02/08/2008.	O.K.	O.K.
7.1.9 In case there is a significant	/10/, /26/	DR	the project was started in March 2011, and the	O.K.	O.K.

gap between the start date of the project activity and the commencement of validation, how was it possible for the project participant to commit funds to the project in advance of receiving a positive validation opinion?	/27/ /28/		site validation was done in Oct. 2011, during site visit, the project is still in construction. This is not a significant gap.		
7.2 Identification of alternatives					
7.2.1 Does the PDD identify and list credible alternatives to the CDM project activity in order to determine the most realistic baseline scenario, unless selected approved methodology prescribes/identifies the baseline scenario and no further analysis is required?	/10/, /15/, /4/	DR	As per the ACM0002 (Version 12.2.0 and VVM, the alternatives to the project is not needed to be identified as the baseline scenario has been prescribed in the methodology itself: The electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by addition of new sources, as reflected in the combined margin (CM) calculations described in the Tool to calculate the emission factor for an electricity system.	O.K.	O.K.
7.2.2 Does the list of alternatives include as one of the options that the project activity is undertaken without being registered as a CDM project activity?	/10/, /15/, /4/	DR	Ditto	O.K.	O.K.
7.2.3 Does the list contain all realistic/credible alternatives that the DOE, on the basis of its local and sectoral knowledge, considers to be viable means of supplying the	/10/, /15/, /4/	DR	Ditto	O.K.	O.K.

outputs or services that are to be supplied by the project activity? <i>Note: All alternatives listed in the selected methodology should be included, as well as those not covered by the methodology.</i>					
7.2.4 Is the exclusion of the alternatives for legal reasons justified? <i>Note: Some alternatives might be illegal, according to the local regulations, but still widely practiced due to lack of enforcement. It should be verified.</i>	/10/, /15/, /4/	DR	Ditto	O.K.	O.K.
7.3 Investment Analysis					
7.3.1 Are all sources of revenues (including savings) have been considered in the PDD and all calculations?	/10/, /11/, /15/	DR, I	Yes, the validation team reviewed the FSR, and other documents, communicated with local stakeholders, and could confirm that all sources of revenues have been considered in the PDD and all calculations.	O.K.	O.K.
7.3.2 Is the type of investment analysis selected correctly in the PDD?	/10/, /11/, /15/	DR	Yes, the benchmark analysis was selected correctly, based on the facts, that the project activity will have revenues and that the baseline scenario (grid consumption) would not be a similar investment project according to the guidelines on the assessment of investment analysis.	O.K.	O.K.
7.3.3 Is the selected financial	/10/, /11/,	DR,	Yes. The selected financial indicator is IRR	O.K.	O.K.

indicator chosen and applied correctly?	/15/ /41/	WWW	based on the Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects. The validation team confirms it is chosen and applied correctly.		
7.3.4 Is the guidance on IRR calculation and assessment correctly applied? <i>Note: Means of validation should be recorded.</i>	/10/, /11/ /15/ /7/, /41/	DR	The validation team has checked the IRR worksheet as per the requirement of Guidelines on the assessment of investment analysis version 04.0 and would like to confirm the following: - The assessment period 21 years consists of 1 years' construction period and 20 years' operational period. The operational period of 20 years is in accordance with 20 year stated in para. 3 of this guidelines. The construction period of 1 years is reasonable considering the actual technical area and capacity scale. The assessment period of 21 years is included in FSR approved by local authority in accordance with para 113 of VVM; - The fair value (residual value of the fixed assets investment) has been included as a cash inflow in the final year of the assessment period in accordance with para. 4 of this guidelines. - The depreciation fee has been deducted in estimating the gross profits (pretax profits). Moreover the selected IRR is a post-tax indicator, the taxation (income tax) has been included as an expense in the item "cash out-flow" of the cash flow statement in accordance with para. 5 of this	GL4	O.K.

			guidelines; - FSR was issued by the independent qualified third party in Dec. 2010, and the project starts in March 2011. The gap is just 3 months, thus the input values is valid and applicable in accordance with para. 6 of this guidelines; - Para. 7 is not applicable for the reason that this project is not a stop- and- recommencement project but a greenfield project according to FSR and its approval; - All formulas used in this IRR worksheet is readable and all relevant cells is viewable and unprotected in accordance with para. 8 of this guidelines; - The PPs choose the calculation approach of project IRR to perform the IRR calculation because the selected benchmark sourced from <i>Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects</i> is prescribed as project IRR. The cost of financing expenditures (i.e. loan repayments and interest) has been excluded in the calculation of cash outflow in the cash flow statement in accordance with para. 9 of this guidelines; - Para. 10 is not applicable for the reason that the approach is selected as "project IRR"; - As the selected benchmark is post-tax project IRR, the actual interest payable is taken into account in the calculation of		
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			income tax in accordance with the requirement in para. 11 of this guidelines. - The Benchmark IRR (project IRR, post-tax) of 8% is applied, which was issued in <i>Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects</i>, a regulation in power industry issued by national authority . It is inaccordance with para. 12 of this guidelines; - Para. 13, 15, 17 are not applicable for the reason that the proposed project is developed by the PP (project owner); - Para. 14, 16, 18 are not applicable for the reason that the benchmark applied is not an internal company benchmark, but a publicly available benchmark issued by the corresponding national authority; - Benchmark approach is considered appropriate for the reason that the alternative (the identified baseline scenario) is the supply of electricity from a power grid, CCPG in accordance with para. 19 of this guidelines; CL4 Please further justify the loan interest rate applied. Since this is the 1st phase of the project in the area, please indicate how many phases will be constructed by the project owner. whether there is a shared cost for all the other phases in the future.		
7.3.5 In case project participants use	/10/, /15/,	DR	Yes, FSR was released in Dec. 2010, the	O.K.	O.K.

values from Feasibility Study Reports (FSR) is it possible to verify that the period between the FSR date and investment decision was reasonably short and FSR values did not change materially?	/20/		board meeting was set out to consider CDM on 15/03/2011, the project started in March 2011. The validation team is able to confirm the interval is reasonably short and it is unlikely that the FSR values would have changed materially.		
7.3.6 Are all the values consistent between FSR and PDD and are inconsistencies properly justified?	/10/, /15/	DR	No, CARs and CL are needed. CAR6 Please justify the calculation of the electricity supplied to the grid accurately and transparently. CL5 Please analyse how the trend of tariff in the whole lifetime will influence the additionality based on the historical and/or official and/or statistical data source. CL6 The PPs should provide a formal announcement that the FSR version provided to DOE for validation is the same version as that provided to local authority for approval.	CAR6 CL5 CL6	O.K.
7.3.7 Were all the values from FSR applicable and valid at the time of the investment decision?	/10/, /15/	DR	CL7 Please conduct cross-checking or other appropriate manner to substantiate the input values from the FSR are valid and applicable at the time of the investment decision in accordance with para 113 (c) of VVM. For the purpose of crosschecking with the total investment, please provide the costs on the main contracts (e.g.: equipment purchase,	CL7	O.K.

			construction contract and so on)		
7.3.8 Is it reasonable to assume that no investment would be made at a rate of return lower than the benchmark by, for example, assessing previous investment decisions by the project participants or some verifiable circumstances that have lead to a change in the benchmark?	/10/, /41/	DR	It is stated in sub-step 2b investment analysis of PDD that "only if the total investment IRR of the project is higher than or equivalent to this benchmark, the proposed project is financially feasible." Having reviewed the section 1.11 of the <i>Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects</i> , the validation team would like to confirm no investment would be made if the IRR is lower than the benchmark.	O.K.	O.K.
7.3.9 Is the Investment Analysis prepared in compliance with the latest version of the "Guidance on the Assessment of Investment Analysis" as provided by the CDM EB?	/10/, /11/	DR	Yes. The latest version of the guidelines on the assessment of investment analysis is version 04. the assessment has been conducted as per the latest version. Please refer to 7.3.4.	Refer to 7.3.4.	O.K.
7.4 Barrier analysis					
7.4.1 Are there any issues addressed in the barrier analysis that have a clear impact on the financial viability of the project activity and that shall be assessed by an investment analysis?	-	-	PPs do not use the barrier analysis.	N/A.	N/A.
7.4.2 Do the listed barriers exist <u>and</u> is their existence substantiated? Note: (a) by independent sources of data such as relevant national	-	-	PPs do not use the barrier analysis.	N/A.	N/A.

legislation, surveys of local conditions and national or international statistics and/or (b) by interviews with relevant individuals: including members of industry associations, government officials or local experts if necessary?					
7.4.3 Would any of the identified barriers prevent the implementation of the project activity but not equally prevent the implementation of the possible alternatives, in particular the implementation of the identified baseline scenario?	-	-	PPs do not use the barrier analysis.	N/A.	N/A.
7.5 Common practice analysis					
7.5.1 If the PPs claim in the PDD that CDM project activity is the “first of its kind”, is it justified?	/10/, Tool	DR	No. The PPs do not claim the project activity is the “first of its kind”. This question is not applicable because it is a wind power project, common in China currently.	O.K.	O.K.
7.5.2 Are the geographical boundaries of the project activity identified correctly?	/10/, Tool	DR	Yes, it is identified correctly that “ only hydropower projects in Shangdong Province which is as the common practice region are analyzed.” CAR7 Please do the common practice analysis according to EB63 Annex 12.	CAR7	O.K.
7.5.3 Does the PDD provide an explanation why this region	/10/, Tool	DR	Ditto	CAR7	O.K.

was selected and deemed more appropriate <u>and</u> is this explanation traceable and reliable?					
7.5.4 Are there similar operational project activities, other than CDM activities, “widely observed and commonly carried out” in the defined region? <i>Note: Use official sources and local and industry expertise.</i>	/10/, Tool	DR	Ditto	GAR7	O.K.
7.5.5 In case there are similar commercially operated project activities, other than CDM activities, already “widely observed and commonly carried out” in the defined region, are there essential distinctions between the CDM project activity and the other similar activities?	/10/, Tool	DR	Ditto	GAR7	O.K.
8. Monitoring					
8.1 Monitoring plan					
8.1.1 Are all parameters required by the selected approved methodology or tool identified <u>and</u> listed in the PDD? Note: not all methodologies indicate monitoring parameters in tabular form or by reference	/10/, /15/, /4/	DR	Yes, all parameters required have been listed in the PDD clearly, i.e. the parameters need to be monitored are net electricity supplied to the grid.	O.K.	O.K.

to the variables used in formulae; Nonetheless, all parameters indicated in the methodology and applicable to the project must be listed in the PDD, omissions due to non-applicability be justified.					
8.1.2 Is the measurement method clearly stated for each value to be monitored and deemed appropriate? Does the monitoring plan record data in the original form as generated, providing QA/QC procedures to be used on the measurement method? <i>Note 1: if the measurement unit is different from the unit to be applied in the methodology, describe the actual measurement and any according conversion method to match the unit used in the methodology.</i> <i>Example: liquid fuels may be monitored as weight or volume. If measured as volume, the measurement method and equipment including the according unit (e.g., liter) shall be described in B.7.1, as well as the conversion into weight units as needed.</i>	/10/, /15/, /4/	DR	Yes, the pp will measure the Egexport and EG inport separately. The net electricity will be obtained from these data. CAR8 Since it is the 1st phase of the project, please indicate the other phases in the monitoring chart and give a clear descriptions on the relations of the meter readings and the EG values, as well as the shared the back-up electricity in the monitoring plan.	CAR8	O.K.

<i>Note 2: Data on invoices / delivery slips may be used for QA/QC purposes, but do not constitute an actual means of monitoring and thus cannot be applied as a source of data.</i>					
8.1.3 Are values of the ex-ante parameters / monitoring parameters selected correctly and conservative in accordance to methodology or tools? See the NOTE in section 3.6.1 above!	/10/, /15/, /4/	DR	Yes, according to ACM0002 version 12.2.0, the ex-ante parameter is the emission factor. Having compared the latest emission factor applied in PDD with that published on the DNA of China's website on 20/12/2010, the validation team is able to confirm it is correctly applicable and conservative. According to the methodology ACM0002 version 12.2.0, the net electricity supplied to the grid is the monitoring parameter. Having reviewed the FSR and its approval separately, the validation team can confirm the estimate provided in PDD is appropriated.	O.K.	O.K.
8.1.4 Is the measurement equipment for each parameter described and deemed appropriate? Are the locations of all measurement equipment clearly identified and consistently described, incl. process flow-charts contained in the PDD?	/10/	DR, I	Yes. The measurement equipment for the parameter, net electricity supplied to the grid is meter, which has been depicted in PDD and deemed appropriate. Refer to CAR8	CAR8	O.K.
8.1.5 Is the measurement accuracy addressed and deemed appropriate?	/10/, /44/	DR	Yes, "no less than 0.5s" is addressed and appropriate in accordance with Technology & Management Regulations for Power Metering Device (DL/T448-2000).	O.K.	O.K.

8.1.6 Are procedures in place on how to deal with erroneous measurements <u>and</u> are the corrective actions identified?	/10/	DR	Yes, in the PDD there are some precodures indicated in place to deal with the erroneous measurement, which are deemed appropriate, and the corrective actions are also identified.	O.K.	O.K.
8.1.7 Is the frequency of measurement identified and deemed appropriate?	/10/, /4/	DR	Yes, "continuous measurement" for meter is addressed and deemed appropriate in accordance with ACM0002.	O.K.	O.K.
8.1.8 Is the monitoring plan documented according to the approved methodology and in a complete and transparent manner?	/10/, /4/	DR	Yes. They are described in the section B.7.2 transparently and clearly.	O.K.	O.K.
8.1.9 Are the sampling, measurement methods and procedures defined? Where applicable, refer to the GENERAL GUIDELINES1 FOR SAMPLING AND SURVEYS FOR SMALL-SCALE CDM PROJECT ACTIVITIES	/10/, /4/	DR	N/A, because 100% of the data should be monitored.	N/A.	N/A.
8.1.10 Are procedures identified for maintenance of monitoring equipment and installations?	/10/	DR	Yes. It has been identified and defined in the section B.7.2 of PDD.	O.K.	O.K.
8.1.11 Are the equipment calibration intervals identified and justified?	/10/, /42/	DR	Yes, the meter will be calibrated annually in accordance with Verification Regulation of Electrical Energy Meters with Electronics (JJG596-1999)	O.K.	O.K.
8.1.12 Are procedures identified for day-to-day records handling (including what records to	/10/	DR	Yes. The procedures are clearly identified and defined in the PDD.	O.K.	O.K.

keep, storage area of records and how to process performance documentation)?					
8.1.13 Are the monitoring arrangements described in the monitoring plan feasible within the project design?	/10/, /15/	DR	Yes. According to the FSR, the monitoring arrangements described in the monitoring plan is feasible. However, refer to CAR8	CAR8	O.K.
8.1.14 Are the means of implementation of the monitoring plan, including the data management and quality assurance and quality control procedures, sufficient to ensure that the emission reductions achieved by / resulting from the project activity can be reported ex post and verified?	/10/	DR	Yes, after reviewed the PDD, the validation team can hereby confirm that the procedures are sufficient to ensure the emission reductions can be reported ex post and verified.	O.K.	O.K.
8.1.15 Do the PPs make provisions for personnel training needs?	/10/	DR	Yes, it has been depicted in B.7.2 of PDD. CAR9 Please provide the CDM monitoring arrangement plan to validation team.	CAR9	O.K.
8.1.16 Is the authority and responsibility of overall project management clearly described?	/10/	DR	Yes. It is clearly indicated in the monitoring plan and the PDD.	O.K.	O.K.
8.1.17 Are procedures identified for emergency preparedness for cases where emergencies can cause unintended emissions?	/10/, /15/	DR	There are no possible emergencies, related to the project activity, which would be able to cause direct unintended emissions because the project activity is a wind power project in	O.K.	O.K.

			renewable energy source category.		
8.1.18 Are procedures identified for review of reported results/data?	/10/	DR	Yes, it has been depicted in B.7.2 of PDD.	O.K.	O.K.
8.1.19 Is the data archiving period for this project activity stated in the PDD and appropriate? <i>Note: All archived monitoring data, required for verification and issuance, should be kept for at least two years after the end of the crediting period or the last issuance of CER.</i>	/10/	DR	Yes, in the monitoring plan it is states that "All data records will be kept for a period of 2 years following the end of the last crediting period." which is in accord with the requirement of the selected methodology.	O.K.	O.K.
8.2 Monitoring of the leakage					
8.2.1 Does the monitoring plan provide for the collection and archiving of all relevant data necessary for determining leakage?	/4/	DR	N/A. As no leakage has to be considered according to the methodology ACM0002 version 12.2.0, no monitoring shall be required.	N/A	N/A
8.2.2 Is the choice of project leakage indicators made according to selected methodology in a reasonable and conservative manner? <i>Note: local knowledge and sectoral expertise shall also be considered.</i>	/4/	DR	N/A	N/A	N/A
8.2.3 Is the measurement method clearly stated and deemed appropriate for each leakage value?	/4/	DR	N/A	N/A	N/A

9. Sustainable development					
9.1 Does the LoA from the Host country DNA contain the confirmation that the proposed CDM project activity contributes to the sustainable development of the host Party?	LoA	DR	Pending due to the lack of LoA, see 1.1.	See 1.1	O.K.
9.2 If PDD indicates any additional environmental benefits of the project, other than GHG emission reductions, were those benefits properly substantiated?	/10/, /15/	DR	Yes, the project creates 20 job opportunity, which can be substantiated by page 1.3.9 of FSR; reduces the emission of SO ₂ , NO _x and soot and improves the living condition of local people which can be substantiated by page 4-6 of FSR.	O.K.	O.K.
10. Stakeholders' consultation and comments					
10.1 Were the stakeholders identified in appropriate and complete manner?	/10/	DR	CL8 Please clarify how many people were involved due to the land expropriation. if possible with providing some evidence. CL9 Please clarify whether the provided questionnaires have covered the people who were impacted directly due to the land expropriation, if not please indicate their comments with providing some evidence. Please scan all the questionnaires to validation team.	CL8 CL9	O.K.
10.2 Are the identified stakeholders plausible?	/10/	DR	See 10.1	CL8 CL9	O.K.

10.3 Does PDD describe the means being used to invite local stakeholder's comments?	/10/	DR	Yes. The means being used to invite local stakeholder's comments is paper questionnaires, conducted in Sep 2010.	O.K.	O.K.
10.4 Were those means appropriate?	/10/	DR	Yes. Those means were appropriate.	O.K.	O.K.
10.5 Was the project presented to the stakeholders in unbiased manner?	/10/	DR	Unclear, Need clarification. CL10 Please clarify how the PPs present the project to the stakeholders in unbiased manner transparently.	CL10	O.K.
10.6 If a stakeholder consultation process is required by regulations/laws in the host country, has the stakeholder consultation process been carried out in accordance with such regulations/laws?	/10/	DR	During EIA process, such a stakeholder's consultation process is required by the EIA legal. And the process and results have been described in the EIA which has been provided by the PP and verified by the validation team. It is in accordance with the regulation. But for CDM application, such a stakeholder's consultation process is not required by the Chinese legal.	O.K.	O.K.
10.7 Is a summary of the stakeholder comments provided in the PDD?	/10/	DR	Yes, it has been depicted in E.2 of PDD.	O.K.	O.K.
10.8 Has due account of any stakeholder comments been taken by PPs and reflected in the PDD?	/10/	DR	Yes. The GSC-PDD has been published directly on the UNFCCC website for a period of 30 days from 10/09~09/10/2011, where no comments were received. Furthermore the comments about the local stakeholders have been taken and reflected in PDD.	O.K.	O.K.

11. Environmental impacts					
11.1 Is the documentation supplied by the PPs regarding environmental impacts relevant and accurately reflected in the PDD?	/10/, /29/ /30/	DR	Yes, the environmental impacts have been reflected in the PDD accurately.	O.K.	O.K.
11.2 Is an environmental impact assessment (EIA) required for the CDM project activity? <i>Note: determine by using a review of relevant legislation and local expertise.</i>	/10/, /29/ /30/	DR	Yes. It is required for the project activity according to the chinese legal.	O.K.	O.K.
11.3 In case an EIA is required, has the EIA has been approved by local authorities and is the outcome accurately reflected in the PDD?	/10/, /29/ /30/	DR	Yes, the EIA was approved on 12/10/2010 by the Environmental Protection Bureau of Qingdao City and the outcome is accurately reflected in the PDD.	O.K.	O.K.
11.4 Does the PDD include a brief description of the environmental effects of the project, including transboundary?	/10/, /29/ /30/	DR	Yes.	O.K.	O.K.
11.5 Are those effects properly addressed in the design of the project activity?	/10/, /29/ /30/	DR	Yes.	O.K.	O.K.
11.6 Does the project comply with environmental legislation in the host country?	/10/, /29/ /30/	DR	Yes.	O.K.	O.K.

Table 2: List of Requests for Corrective Action (CAR) and Clarification (CL)				
CAR/CL No.	Observation (CAR/CL)	Reference	Summary of project owner response	Validation team conclusion
CAR1	CAR1 Please submit the LoA of the host country for validation when available.	1.1~1.7 2.1~2.2	The LoA from China's DNA is impending, it will be provided as soon as it is received.	LoA of the host country is provided, this CAR is closed.
CAR2	CAR2 Please describe the detailed project location in A.4.1.4. From the map, we can not see the Longxin as a location. Please demonstrate the technical training arrangement of the Operation and Maintenance of the plant. Please submit the layout of the project and the project schedule. Please provide the Project IRR sheet and ER sheet to Validation team.	4.1	Longxin with the meaning of propitiousness was just used by the PP to nominate the proposed project, but not the geographic name. The proposed project is located at Dianzi Town and Liyuan Town, Pingdu City, Shandong Province, P. R. China. A clear map is applied in the revised PDD. The technical training arrangement of the Operation and Maintenance, the layout of the project and the project schedule and the Project IRR sheet and ER sheet are provided.	Yes, the revised map can show the actual location of the project; the layout and the geographic coordinates are clearly shown in the PDD and the provided document; and the project schedule was provided and it meets site construction status during OSV. The Operation and Maintenance Procedures are provided too; The project IRR and ER sheet are provided too. And the annual training plan for year 2012 is provided, So, this CAR is closed.
CAR3	CAR3 PP is requested to indicate the type of turbines in A.4.3 of PDD, and fill the key events on the timeline in table in the B.5 in the PDD.	4.1	The type of turbines was added in A.4.3 of PDD in line with the technical parameters in the turbines purchase contract. The key events of notification	Follow-up 1: The construction contract signed on May 19 2011 should be in

	Please show to validation team the complete version of the FSR (incl. Cover and second page) which was to submit to DRC for approval. Please submit the land use approval for the project.		to DNA and EB are also appended to the timeline. Please refer to the revised PDD. The complete version of the FSR submitted to DRC and the land use approval are provided. 2nd Response from PP: The signature of construction contract was added into the time table. The right of land use was approved by the local government. But the Land use approval is unavailable at present and pre-trial opinions on pre-construction land issued by Land and Natural Resources Departments of Shangdong province was provided.	the time table. The Land use approval is not provided, pp has just provided the land compensation agreement. Conclusion: Yes, the land-use pre-approval is provided and the construction contract is listed in the project timetable. This CAR is closed.
CAR4	CAR4 The format of the geographical coordinates should meet the following requirement: – All GPS coordinates indicated in the PDD should be presented in the decimal format with +/- sign only. Coordinates with multiple types of specifiers will not be accepted. – All GPS coordinates should have a precision of 4 (four) decimals. Lower precision will not be accepted. Higher precision will have to be rounded up, otherwise will not be accepted. - GPS coordinates for the project implementation area (ex. square, triangle, polygon, etc.) in the PDD; - Single set of GPS coordinates in the PDD, which would depict an approximate geometrical middle/center point of the area of the implementation; - A table with a set of coordinates for each implementation point.	4.1	Please refer to the revised PDD, the geographical coordinates has been presented in four decimals. The geographical coordinates of the proposed project is from 119.8827E to 119.9503E, and from 36.8157N to 37.8915N. The detailed coordinates of the 33 wind turbines are provided.	Yes, the geographical coordinates of the project is provided, and validation team verified it correct by using google-earth, this CAR is closed.

CAR5	CAR5 Please depict if there are some risks and uncertainties, which can influence the emission reduction estimates. If yes, please identify and address it in PDD.	5.5.5	The feasibility study report of the proposed project was developed by a qualified entity, Shandong Industrial Design Institute Co., Ltd. The estimated output of the proposed project was caculated by special software based on the historical data of the local wind resource which is reasonable. According the methodology ACM0002(vesion12.2.0), there is no fossile fuel consumption involved and the project emmission for the proposed project is zero. Meanwhile no leakage emissions are considered. So there are no risk and uncertaintie influencing the emission reduction estimates.	Yes, the project was designed by national qualified design institute by more than 10 years historical data and the baseline emission scenario is correctly demonstrated, there is no leakage was detected according to the methodology, hence, there is no significant risk and uncertainties to influence the emission reduction estimation, this CAR is closed.
CAR6	CAR6 Please justify the calculation of the electricity supplied to the grid accurately and transparently.	7.3.6	The annual theoretical power generation sourced from the FSR is 139,159 MWh which is calculated by special software based on historical data of wind resource. Considering the correction of 71.16%, the net annual power generation is calculated as $139,159 * 71.16\%$ which equals to 99,026 MWh. Thus the electricity supplied to the grid is accurately and transparently calculated. 2nd Response from PP: It involves the number of the decimal places in the calculation of the parameter of 71.16%, and the outcome will be not equivalent if each factor is multiplied directly. A statement is	Follow-up 1: PP is requested to clarify how does the 71.16% come, since the validation team gets 71.90% by calculating the coefficients in table 6-5 of FSR. Conclusion: A clarification was provided by the original design institute, this can support the calculation of the electricity supplied to

			supplied by Shandong Industrial Design Institute Co., Ltd.	the grid in the FSR, this CAR is closed.
CAR7	CAR7 Please do the common practice analysis according to EB63 Annex 12.	7.5.2~7.5.5	The validation of the proposed project is started from 10 September 2011, and at that time the available tool applied to analyse the common practice is "Tool for the Demonstration and Assessment of Additionality (version 05.2.1)". In EB meeting 65 held from 21 to 25 November 2011, "Guidelines on Common Practice (version 01)" (EB63 Annex 12) was integrated into the "Tool for the Demonstration and Assessment of Additionality" and the Tool was updated to version 06.0.0. However, according to the report of EB65, <i>"For all revised methodologies and tools that were approved by the Board at this meeting, the DOEs may upload not later than 25 July 2012 (24:00 GMT) for registration the project design documents (PDDs) of project activities in which the previous version of an approved methodology or an approved tool has been applied"</i>, therefore, "Tool for the Demonstration and Assessment of Additionality (version 05.2.1)" is still valid before 25 July 2012, it is not necessary to apply the new guidelines of common practice to The proposed project. 2nd Response from PP: The scanned copies for footnote 17 was	Follow-up 1: Please provide the scanned copies for footnote 17 to validation team. Any wind power projects were developed from 2003 to 2006 in Shandong Province? in addition, 15MW is defined as the bottom range, and any upper range for the scale comparison? Conclusion: Yes, the scanned copy of footnote 17 is provided; the wind power projects developed from 2003 to 2006 in Shandong Province, which were included in the "Statistic of installed capacity of China's Wind Farms in 2007". The explanation of the project scale is acceptable. The common practice

			provided. There are wind power projects developed from 2003 to 2006 in Shandong Province, which were included in the "Statistic of installed capacity of China's Wind Farms in 2007". Wind farm within 15MW is defined as small scale CDM project in CDM guidance and the bottom range for comparison with the proposed project is defined to 15MW.	is coorrectly done, this CAR is closed.
CAR8	CAR8 Since it is the 1st phase of the project, please indicate the other phases in the monitoring chart and give a clear descriptions on the relations of the meter readings and the EG values, as well as the shared the back-up electricity in the monitoring plan.	8.1.2 8.1.4 8.1.13	There will be two projects constructed including the proposed project, another project which refers to Phase II Project is still under preliminary design stage and developed by the same project owner with the proposed project. The electricity supplied to the grid by the proposed project and Phase II Project will be delivered to the grid through the 110kV transmission line. The proposed project will share the 110 transmission line with Phase II Project. Meter M1 installed on the Tangtian Substation as the main meter will be used to measure the electricity supplied by the proposed project and Phase II project. Meter M2 and M3 installed at the on-site substation will be used to measure the electricity supplied by the proposed project and Phase II project respectively. Sales receipts on the electricity supplied to the grid by Phase II Project will be issued according to meter reading from M3. Sales receipts	Yes, the explanation is clear, the meters locations and the EG values are demonstrated clearly in the monitoring plan of revised PDD, this CAR is closed.

			on the electricity supplied to the grid by the project activity will be issued according to meter reading from M1 minus meter reading from M3. The reading of meter (M2) is carried out by the project owner for cross check. The precision of the meter M1, M2 and M3 will be not lower than 0.5s. A spare 10 kV line is used to import electricity from the grid in emergency and the electricity can be measured by a meter (M4). The net electricity supplied to the grid by the proposed project (EG_{facility,y}) is calculated as followed: $EG_{facility,y} = EG_{output,y} - EG_{another,y} - EG_{import,y} - EG_{im-spare,y}$ EG_{output,y}: Total electricity supplied to the Grid by the proposed project and Phase II project in the year y; EG_{import,y}: Total electricity imported from the Grid by the proposed project and Phase II project in the year y; EG_{im-spare,y}: Electricity imported from the Grid through a spare 10 kV line in the year y; EG_{another,y}: Electricity supplied to the Grid by the Phase II Project in the year y.	
CAR9	CAR9 Please provide the CDM monitoring Manual and training arrangement to validation team, if any.	8.1.19	CDM manual monitoring arrangement is provided. 2nd Response from PP:	Yes, the CDM monitoring Manual was provide to validation

			CDM monitoring training plan was provided	team, however, please provide the CDM monitoring training plan for the project to validation team.
CL1	CL1 Any loan contract signed, if yes, please provide scanned copy to validation team. Please show evidence that the project is not a project of ODA involved in. Please provide the current organization chart of the Company. Please provide the CDM consulting contract of the project to validation team.	4.1	According to the FSR, total investment of the Project is 571.86 million yuan which consist of capital of 180.62 million yuan (30% of the total investment) and loan of 361.24 million yuan (70% of the total investment) In the license of the project owner, the capital is 180.62 million yuan which is the same with the value in the FSR. 70% of the total investment will come from the loan, the loan contract will be provided when be signed, therefore, there is no need that the proposed project be supported by ODA. Organization chart of the project owner and CDM consulting contract signed with Easy Carbon Consultancy Co. Ltd. are provided. 2nd Response from PP: There is no available asset evaluation report for the project and a statement from PP on ODA was supplied.	The organization chart and CDM consulting contract were provided, and instead of bank loan money, the PP confirmed that the capital for the project were come from internal leasing in the China Resource (Group) Ltd, Yes, A statement from PP on ODA issue was provided to validation team, and no ODA involved in the project is confirmed. This CL is closed.
CL2	CL2 Please clarify whether there is any possible GHG emission source or sink occurring as a result of project activity, which have not been accounted for under the selected methodology and are expected to contribute more than 1% of the overall expected average annual emission reductions. If yes, please consider it as per para	5.2.5	According to the response to CAR8, there is a spare 10 kV line used to import electricity from the grid in emergency. The electricity imported from the grid by the proposed project and Phase II Project is conservatively included when the EGfacility,y is calculated. The details please refer to	Yes, as confirmed by the Project owner, there is a spare 10kV transmission line for the import of electricity from the grid. No other emission sources were identified by the

	77 of VVM.		the revised PDD.	validation team, this CL is closed.
CL3	CL3 Why the submitted contracts did not include the construction of powerhouse or/and admin house?	7.1.2	It is shown that construction of admin house is included in the Construction Contract of Road, foundation, platform, turbine installation, lightning protection and substation. The related pages of the contract were scan and provided again.	Yes, the newly scanned contract includes the admin house of the project, this CL is closed.
CL4	CL4 Please further justify the loan interest rate applied; Since this is the 1st phase of the project in the area, please indicate how many phases will be constructed by the project owner. whether there is a shared cost for all the other phases in the future.	7.3.4	The FSR is completed on 28/12/2010 and in which the interest rate for longterm and shortterm loan is 6.40% and 5.81% respectively, the interest rate guided by The People's Bank of China-Chinese central bank is refered to when the FSR developed. It can be cross checked by the data published on the website (http://www.pbc.gov.cn/publish/zhengcehuobisi/627/index.html). The same interest rates were applied in the IRR spreadsheet and PDD. The project owner plan to construct two phases. The costs of the shared main transformer and electrical equipment are undertaken by the proposed project and will not be contained in the investment of the Phase II project.	Yes, the long term short term interest rates are clearly shown by the linkage, the FSR used the correct one for the project feasibility estimation; For the shared costs for the two phases, the explanation can meet the content in the FSR of the phase I project, this CL is closed.
CL5	CL5 Please analyse how the trend of tariff in the whole lifetime will influence the additionality based on the historical and/or official and/or statistical data source.	7.3.6	According to the historical tariffs in Shandong Province, the tariffs of the wind power projects are in the range of .61-0.76 CNY/kWh (Inc. VAT). However, since the Power Sector Reform in March 2002, the tariffs of the wind power projects is stable on 0.61 CNY/kWh (Inc. VAT). On 23/07/2008, the first tariff notification (Document	As the wind power tariffs in Shangdong Province is very stable, and the validation team verified the mentioned government electricity tariff policies and can confirm that the tariff will have no significant

			Fagaijiage[2008]1876) regarding to wind power projects in Shandong province was issued by National Development and Reform Commission (NDRC). In this notification, the tariff of 10 wind power projects in Shandong province was approved to be 0.61 CNY/kWh (incl. VAT). While referring to the Tariff Approval for Wind Power Project [Fa Gai Jia Ge (2009) No.1906] issued by NDRC in 2009, the guiding tariff of wind power plants in Shandong province is 0.61CNY/kWh (Inc. VAT). The tariff of 0.61Yuan/kWh was used in the FSR (and thus also used in the PDD), which was based on the wind power on-grid tariff policy released by NDRC on 20/07/2009. The staticstic on the approved tariffs of all Shandong wind farm projects since the Power Sector Reform in March 2002 shows that all projects from then on have received the same tariff: 0.61 CNY/kWh. In addition, according to the "Information Note on the Highest Tariffs Applied by the Executive Board in its decisions on Registration of Projects in the People's Republic of China"(version 02), the highest tariff in Shandong Province is 0.61 CNY/kWh. Therefor the tariff in Shandong is deemed to be stable on 0.61Yuan/kWh and have little influence on the additionality to the proposed project. Details are presented in the revised PDD.	influences on the additionality of the project, this CL is closed.
CL6	CL6 The PPs should provide an formal	7.3.6	The FSR version provided to DOE for validation is the same version with the	Yes, from the Expert

	announcement that the FSR version provided to DOE for validation is the same version as that provided to local authority for approval.		FSR provided to local authority for approval. The evidence document from Shandong Engineering consulting Institution authorized by the project owner to organized the expert review is provided. The document is show that parameters such as the total investment, annual power supply, tariff and so on in the FSR is in line with the version provided to DOE for validation.	Evaluation Report for the Longxin Wind Farm Project can show the same parameters are mentioned as those of in the FSR provided to the validation team. Hence this CL is closed.
CL7	CL7 Please conduct cross-checking or other appropriate manner to substantiate the input values from the FSR are valid and applicable at the time of the investment decision in accordance with para 113 (c) of VVM. For the purpose of crosschecking with the total investment, please provide the costs on the main contracts (e.g.: equipment purchase, construction contract and so on)	7.3.7	The annual electricity output in the FSR is calculated based on the historical wind resources assessment and professional software. The FSR of the proposed project was evaluated and approved by Shandong Development and Reform Commission, including the annual electricity output assessment. In the FSR, the annual theoretical power generation is 139,159 MWh. Considering the correction of 71.16%, the net annual power generation is 99,026 MWh. The wind data from FSR shows that wind resources at the project site keep stable. The annual electricity generation in the FSR is estimated based on the long term wind data (from 1980 to 2009) analysis, therefore, the annual electricity output is reasonable and applicable. The annual O&M cost in FSR is estimated according to relative national standards and regulations. Considering all wind farm projects in Shandong Province which have been registered, the highest percentage of annual O&M	Follow-up 1: What are the contracts have not been signed so far, since the contracted amount stands for 90.11% total static investment. Conclusion: Yes, the validation verified the contract list and confirmed all the contracts are real and the data on the contracts are transparent, hence this CL is closed.

			cost in the total investment cost is 4.81%, and the proposed project is 2.19%, which is lower than 4.81%. Thus it is demonstrated that the estimated annual O&M cost is reasonably applied in the financial analysis. Referring to the Tariff Approval for Wind Power Project [Fa Gai Jia Ge (2009) No.1906] issued by NDRC in 2009, the guiding tariff of wind power plants in Shandong province is 0.61CNY/kWh (Inc. VAT).Therefore, the tariff of 0.61Yuan/kWh was used in the FSR (and thus also used in the PDD), which was the latest available on-grid tariff at the time of the preparation of the FSR. Besides, Considering “Information note on the highest tariffs applied by the Executive Board in its decisions on registration of projects in the People’s Republic of China” (version 02), The highest tariff in Shandong Province is 0.61 Yuan/kWh, Therefore, the tariff applied is reasonable and sound. The comparison between the contract value and the estimated value has been provided in PDD. The contract value has been 90.11% of the total statistic investment in PDD even the project has not completed construction. The relevant evidence for crosschecking with the total investment has been provided to DOE.	
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			2nd Response from PP: There are some other contracts with small amount, such as survey and design on power transmission project, some small electrical equipments purchase. Considering the ammount of these contracts, the actual costs will exceed total static investment in the FSR.	
CL8	CL8 Please clarify how many people were involved due to the land expropriation. if possible with providing some evidence.	10.1 10.2	According the EIA and FSR, the land where the proposed project located belongs to uncultivated land and no people were involved in the land expropriation.	Yes, the answer is in line with the comments from the stakeholders from the OSV, there is no land expropriation for this project, this CL is closed.
CL9	CL9 Please clarify whether the provided questionnaires have covered the people who were impacted directly due to the land expropriation, if not please indicate their comments with providing some evidence. Please scan all the questionnaires to validation team.	10.1 10.2	No people were involved in the land expropriation. The questionnaires were carried out at Liyuan Town, Mencun town and Dianzi town which locate around the site of the proposed project. The questionnaires are provided to the validation team.	Since the PP randomly raised the stakeholder survey by questionnaire distribution in Mencun Town and Dianzi Town of Liyuan Town. This CL is closed.
CL10	CL10 Please clarify how the PPs present the project to the stakeholders in unbiased manner transparently.	10.5	The project participants have carried out a public survey on the project in the format of questionnaires. The survey was conducted through distributing and collecting responses to a questionnaire. The questionnaires were delivered in date 20 March 2011 to 50 villagers nearby the project area and 49 responses were collected at the same day. The respondents mainly included aquiculture farmers and local residents inhabiting in the nearby villages. The	Yes, the validation team confirms that a public notice have been raised in the website of Huarun New Energy before the stakeholder comments survey. After that, 50 questionnaires had been distributed and 49 returned. The validation team can confirm that the public

			respondents included people of	survey was done in unbiased manner transparently. This CL is closed.
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Appendix B

Certificates of Competence

Qualification

Huang, Minglong /

Emission Trading

United Nations Framework Convention on Climate Change

Auditor No. :
(AuditorenRegNr)

Appointed:
(Zugelassen)

☒ ja

Qualification Level:
(Qualifikationsstufe)

External:
(Externer)

☐ ja

Add. reviewer: ☐ yes
(Zus-tzlicher Prüfer)

EAC Scopes:
(EAC Branchen)

CDM 01 – Energy industries (renewable – / non-renewable sources)
CDM 05 – Chemical industry
CDM 11 – Fugitive emissions from production and consumption of halocarbons and sulphur hexafluoride
CDM 12 – Solvents use

Add. qualification:
(zus. Qualifikation)

First Appointment:
(Erstberufung)

2009-12-28

Valid to:
(Gültig bis)

2012-12-27

Remarks:

CDM 01 valid for TA1.2 – Renewable Energies
CDM 5.1 / 11.1 / 12.1 – Chemical process industries

Languages:

Chinese
Chinese simplified
English
Cantonese
French

Experience Exchange

Date

Location

Remarks

Accreditation

Monitoring

Latest Monitoring:
(letzte Beurteilung)

Next
Monitoring:
(n-chste)

Beurteilung)

Remarks:

[View](#) / [Edit Monitoring](#)

History of scope allocation

Date:
Change:
By:
Reason:

Date: 2010-03-30
Change: EAC CDM, CDM, CDM added
By: Manfred Brinkmann
Reason: Scope 1 / renewable energies (except biomass)

History

Created:	2010-01-13 14:23:26	Minglong Huang/Sz/Chn/TUV
Modified:	2011-01-13 15:46:27 ZE9	Manfred Brinkmann/Jpn/TUV
	2011-01-13 15:45:40 ZE9	Manfred Brinkmann/Jpn/TUV
	2011-01-04 15:27:57 ZE9	Manfred Brinkmann/Jpn/TUV
	2011-01-04 15:27:42 ZE9	Manfred Brinkmann/Jpn/TUV
	2011-01-04 15:26:53 ZE9	Manfred Brinkmann/Jpn/TUV
	2011-01-04 15:25:18 ZE9	Manfred Brinkmann/Jpn/TUV
	2011-01-04 15:24:11 ZE9	Manfred Brinkmann/Jpn/TUV
	2010-03-30 01:45:18 ZE9	Manfred Brinkmann/Jpn/TUV

Qualification

Li, Lixin /

Emission Trading

United Nations Framework Convention on Climate Change

Auditor No.:

(AuditorenRegNr)

Appointed:

(Zugelassen)

☒ ja

Qualification Level:

(Qualifikationsstufe)

Lead Auditor

External:

(Externer)

☐ ja

Add. reviewer:

(Zusätzlicher Prüfer)

☒ yes

EAC Scopes:

(EAC Branchen)

CDM 01 - Energy industries (renewable - / non-renewable sources)

CDM 03 - Energy demand

CDM 02 - Energy distribution

CDM 04 - Manufacturing industries

Add. qualification:

(zus. Qualifikation)

First Appointment:

(Erstberufung)

09/06/2010

Valid to:

(Gültig bis)

09/05/2013

Remarks:

Appointed as Technical Reviewer for TA 1.1, 1.2, 2.1, 2.2, 3.1
TA 4.5

Languages:

Experience Exchange

Date

Location

Remarks

Accreditation(s)

2010-12-21 Beijing

GC CDM Auditor Experience Exchange, Beijing, 2010-12-21 to 23

United Nations Framework Convention on Climate Change

Monitoring

Latest Monitoring:

(letzte Beurteilung)

Next Monitoring:

(nächste Beurteilung)

Remarks:

History of scope allocation

Date: 2012-03-10
Change: EAC CDM, CDM added
By: Praveen Urs
Reason:

Date:
Change:
By:
Reason:

Date:
Change:
By:
Reason:

Date: 2010-11-08
Change: EAC CDM, CDM added
By: Manfred Brinkmann
Reason: Appointed as Technical Reviewer for

History

Created:	08/13/2010 11:09:24 AM	Lixin Li/Bj/Chn/TUV
Modified:	07/06/2012 04:55:01 PM	Praveen Urs/Chn/TUV
	03/10/2012 08:33:44 PM	Praveen Urs/Chn/TUV
	02/12/2012 06:12:39 PM	Praveen Urs/Chn/TUV
	11/15/2010 04:02:03 PM	
	11/15/2010 04:01:56 PM	
	11/08/2010 09:36:09 AM ZE9	
	11/08/2010 09:28:17 AM ZE9	
	11/08/2010 09:28:07 AM ZE9	
	11/08/2010 09:27:39 AM ZE9	
	08/13/2010 11:09:41 AM	